IBM HR Analytics Employee Attrition: A Statistical Exploration

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Abstract

Employee attrition is a significant problem and impacts the stability, growth, or most importantly the competitiveness of an organization. At a time of rapid technology change and evolving work styles, we need to understand the driving forces that are causing employees to walk out. The focus of the project was to find out what leads an employee to leave the organization by understanding how all factors are related to one another. This study was guided by the following research questions: 1) What are some numerical and categorical factors that influence attrition? 2) How do these factors interact, and what can a company change in order to reduce turnover?

In this study, we investigate these questions using an Exploratory Data Analysis (EDA) and statistical methods, including the ANOVA and Chi-Square tests, to evaluate which variables are significant. We analysed a dataset that includes numeric as well as categorical features in the context of employee demographics, job roles, compensation, and satisfaction levels. The ANOVA test analyses numeric features on attrition, while the Chi-Square test examines categorical features and employee turnover. Visualizations are used to show the results, making it easy for someone to see which factors were most responsible for attrition.

The research raises important ethical, legal, and social concerns. As for the moral, it magnifies transparency and maturity in doing fair justice towards employees. All efforts to alleviate attrition pressures must be made in a fair and equitable manner, without recourse to prejudice or bias. In explaining how one should proceed legally, the study indicates that retention strategies must align with labour laws and regulations. On the social side, the research reveals that attracting and retaining requires being mindful of each individual's needs as a professional in tandem with wanting family free time.

The outcomes of this project will be valuable to human resource professionals, organizational leaders, and policymakers seeking to enhance employee retention. This could help inform HR strategies that are designed to pinpoint the predictors of attrition and stop early churn in studies. By employing these strategies, organizations can reduce the direct costs of turnover such as the expense in recruiting and training new employees or losing valuable institutional knowledge. In the end, this work contributes to advancing discussions surrounding workforce management by providing evidence of trends that can provide recommendations for how to establish a more stable, happy, and productive workplace.

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CHAPTER 1. Introduction

Employee turnover poses significant challenges for organizations with increased expense in recruiting a new employee, wastage of productivity, and reduced morale inside the workplace. The IBM HR Analytics Employee Attrition & Performance dataset is studied to understand and analyse the factors driving employee turnover in this study. This research applied Exploratory Data Analysis (EDA) and ANOVA, Chi-Square statistical tests were used to test the significance of demographic characteristics, job-related factors, and organizational variables in relation to attrition. The results will be even more valuable since they allow organizations to identify why their employees are leaving and develop specific strategies to increase employee retention rates. These steps lead to a lower turnover rate, greater employee retention, and simply better organizational performance.

1.1 Background And Context Of The Study

Organizations face a major problem in the form of employee attrition directly linked to productivity, morale, and financial stability. High staff turnover leads to higher costs associated with hiring and training new employees, broken business processes, and loss of organizational knowledge. The IBM HR Analytics Employee Attrition & Performance dataset is an invaluable option to perform this analysis of what factors contribute most to employee attrition and performance. This study examines the main inferences inducing turnover using Exploratory Data Analysis (EDA) so that firms can take early strides with the aim of facilitating improved employee retention and overall company performance.

1.2 Problem Statement And Research Questions

High employee attrition within organizations particularly when it surpasses industry benchmarks is a red flag that may be pointing towards more systemic issues like discontent, no career progression, or work-life balance. These factors lead to an ongoing cost of rehiring and have a depleting effect on the general atmosphere in the office.

RESEARCH OBJECTIVE: The problem posed for scrutiny in this study is analysing the essential causes of employee turnover & retention and then attempt to reveal wonderful interventions helping keep down each pipe down lengthiness, which renders hearty a contractor accomplishment.

The research will address the following questions:

I. What is the current attrition rate in the organization, and how does it compare to industry standards?
II. Which demographic factors are associated with higher attrition rates?
III. How do job-related factors, such as job role, department, and performance rating, impact attrition rates?
IV. How does the length of service in the organization and in the current role affect attrition rates?
V. What is the impact of compensation and benefits on employee attrition?
VI. Are there any significant differences in attrition rates between employees who have been promoted and those who have not?

1.3 Objectives And Significance Of The Research

The aim of this research is to do a detailed Exploratory Data Analysis (EDA) on the IBM HR Analytics Employee Attrition & Performance dataset, which reveals insights and trends with respect to employee turnover, for only understanding purposes. This research, therefore, strives to identify the elements most determinately linked with turnover and thus inform organizations as to where they

should direct their efforts in order to optimize retention strategies, minimize attrition costs, and elevate employee morale across a department or organization of employees at large.

This research is important as it can assist companies in identifying drivers of attrition and taking corrective measures based on data in order to retain their employees more effectively. These insights can help organizations formulate specific programs such as the enhancement of pay and benefits packages, upgrading career growth avenues, or introducing work-life balance initiatives, creating a more committed and loyal workforce.

1.4 Scope And Limitations

The scope of this study is confined to the IBM HR Analytics Employee Attrition & Performance dataset. The research focuses on analysing various demographic, job-related, and organizational factors to determine their impact on employee attrition. The study employs statistical methods such as ANOVA and Chi-Square tests to evaluate the significance of both numerical and categorical variables.

However, the research has several limitations. The dataset may not capture all the factors influencing employee attrition, such as external economic conditions or personal reasons for leaving. Additionally, the analysis is based on historical data, which may not fully represent current or future trends. The findings are specific to the dataset used and may not be generalizable across all industries or organizations. Despite these limitations, the study provides valuable insights that can guide organizational strategies for reducing employee attrition.

CHAPTER 2. Literature Review

In this Literature review, we aim at completing a thorough analysis of the current research and scholarly texts on the selected subject. To do this, our purpose is to put together findings and critiques from different academic sources including peer reviewed articles and other relevant publications as such. Our review also intends to bring out the major patterns about the material examined as well as areas which remain unknown in existing study. In addition, this synthesis is an important point for further researches since it has some important points that will help us move forward in our studies in the discipline.

This work leverages machine learning (ML) for predicting employee attrition, addressing a crucial aspect of HR analytics driven by the need for data-driven decision-making. Factors such as job satisfaction, organizational commitment, and engagement, as highlighted by Haus Knecht et al. (2009) and Hom et al. (2017), significantly influence employee turnover. Various ML algorithms, including decision trees, logistic regression, and support vector machines (SVM), have been applied to predict attrition. Studies by Khera and Divya (2019) suggest ensemble methods are superior, while Sharma and Goyal (2018) found SVMs highly accurate. In this study, the Gaussian Naïve Bayes classifier effectively predicted attrition with a recall rate of 0.54 and a false negative rate of 4.5%, aligning with findings by Mishra and Ranjan (2018) on Naïve Bayes' efficacy. Comparative analyses, such as Chaitanya and Srivastava (2020), indicate random forests often outperform other models due to their ability to handle non-linear relationships. Nonetheless, data quality issues and ethical concerns, as discussed by Cappelli (2019), pose significant challenges. Future research should enhance data quality, explore advanced algorithms, and address ethical considerations to improve prediction accuracy and fairness in HR practices. **(Fallucchi, 2020)**

This work examines the pressing issue of employee attrition and how it impacts organizations' productivity and growth. The authors employ machine learning techniques, particularly Random Forest, to predict employee turnover using HR analytics data from IBM Analytics. The dataset includes 35 features and 1470 samples, with the Random Forest Classifier being optimized using the SMOTE mechanism to address class imbalance. Despite improvements in training metrics post-SMOTE, the

validation metrics, especially sensitivity, showed minimal enhancement. The study identifies critical factors influencing employee attrition, providing valuable insights for top management to make informed decisions about workforce engagement. Previous studies have utilized various machine learning models such as Naïve Bayes, Logistic Regression, and Support Vector Machines to address similar issues. However, this research highlights the superiority of Random Forest in achieving higher accuracy and robustness against overfitting. The findings suggest that incorporating such predictive models can aid organizations in proactive planning and better recruitment strategies. Future research is suggested to further refine the model and reduce the prediction error margin, thereby enhancing its practical applicability in organizational settings. **(Krishna, 2022)**

This work focuses on employee attrition prediction using advanced predictive analytics techniques. It proposes a comprehensive approach integrating machine learning, deep learning, and ensemble methods to predict and interpret employee turnover. The study emphasizes feature selection to identify the most influential factors impacting attrition, such as job satisfaction, business travel, and rewards. Methodologically, it employs diverse datasets—large-scale simulated datasets from Kaggle and IBM, and a smaller real-world dataset—to validate model performance across different settings. The findings highlight the Voting Classifier (VC) as the top-performing model, showcasing its superiority in accuracy for both simulated and real datasets. Interpretation of results underscores the significance of business travel as an often-overlooked factor in attrition, suggesting novel retention strategies for HR managers. Limitations include the need for dynamic feature incorporation and handling of unbalanced data in future research. Overall, this study contributes a refined predictive model tailored for employee attrition, enhancing HR decision-making capabilities through nuanced data-driven insights. **(Yahia, 2021)**

This work explores the critical issue of employee attrition prediction using data mining techniques, focusing on the IBM HR dataset. The study underscores the detrimental impact of employee turnover on organizational stability and financial performance, particularly when skilled personnel depart. The motivation stems from organizations' increasing need to pre-emptively manage attrition by leveraging predictive analytics. Traditional HR methods often fall short in accurately forecasting attrition, prompting the adoption of advanced data mining approaches. The research evaluates eleven classification algorithms, including SVM, Random Forest, and Logistic Regression, to predict attrition rates. Unlike previous studies, which primarily focus on accuracy alone, this study comprehensively assesses multiple performance metrics such as sensitivity, specificity, F-measure, and Area Under Curve (AUC). Findings highlight the effectiveness of these methods in predicting attrition, with Logistic Regression, AdaBoost, and LDA showing promising results in achieving high accuracy. Existing literature, summarized in the study, reveals varied approaches to predicting employee turnover, emphasizing different algorithms and datasets. However, a standardized evaluation framework integrating multiple metrics remains underexplored. This study addresses this gap by providing a detailed comparison and performance evaluation of diverse classification methods, offering valuable insights for HR departments seeking proactive attrition management strategies. **(Ozdemir, 2020)**

This work reviews recent advancements in predicting employee attrition using machine learning techniques. It explores various methodologies applied to datasets like the IBM HR Employee Attrition dataset, highlighting approaches such as SVM, LR, DTC, and ETC. Previous studies emphasize data pre-processing techniques like SMOTE for dataset balancing and feature engineering to enhance model accuracy. Exploration of factors such as monthly income, job level, and age through Employee Exploratory Data Analysis (EEDA) underscores their impact on attrition rates. Comparative analyses reveal ETC as achieving the highest accuracy of 93%, outperforming SVM (87%), LR (72%), and DTC (83%). Validation through k-fold cross-validation and ROC curve analysis consistently supports the efficacy of these models. This review identifies a shift towards applying deep learning methods for future research, aiming to expand dataset features and improve prediction accuracy in employee attrition studies. **(Raza, 2022)**

This work conducts a comprehensive evaluation of machine learning models—J48, SVM, and ANN—using the IBM HR Analytics Employee Attrition and Performance dataset. The study emphasizes the impact of feature selection and parameter tuning on model performance. Through rigorous experimentation and 10-fold cross-validation, SVM emerges as the top performer, achieving an accuracy of 88.87% after fine-tuning with a C=10 parameter under the PUK kernel. This finding underscores SVM's robustness in handling high-dimensional data and its efficacy in predicting employee attrition. In contrast, Decision Trees (DT) and Artificial Neural Networks (ANN) exhibit lower accuracies of 84.40% and 87.38%, respectively. The research highlights the significance of pre-processing steps in optimizing model outcomes, addressing challenges like data imbalance and dataset size. Future directions include feature importance analysis and further optimization through hyperparameter tuning techniques. Overall, this study contributes valuable insights into optimizing predictive models for employee attrition prediction in organizational settings. **(Mansor, et al., 2021)**

This work proposes a three-stage framework for predicting employee attrition using machine learning, emphasizing the critical roles of pre-processing, processing, and post-processing stages. In the pre-processing stage, feature selection is conducted using the "max-out" algorithm, which efficiently reduces dimensionality while accommodating both binary and continuous features in the dataset. This method ensures robust selection without excessive computational overhead. Moving to the processing stage, a logistic regression model is trained on the selected features, focusing on the significance of coefficients to pinpoint influential factors driving attrition. Model performance is rigorously evaluated using metrics such as accuracy, precision, recall, and F1-score, demonstrating enhanced efficacy through optimized feature selection strategies. In the final post-processing stage, parameter confidence is analysed using bootstrapping techniques to validate model stability and generalizability across diverse datasets. By integrating comprehensive feature selection, thorough performance evaluation, and robust parameter validation, this approach addresses existing gaps in attrition prediction literature, offering a more accurate and applicable tool for human resource management strategies. **(Najafi-Zangeneh, et al., 2021)**

This work explores the critical issue of employee attrition in the context of business process outsourcing (B.P.O.) sectors, emphasizing the substantial impacts on organizational performance and financial costs associated with losing skilled employees. Employee turnover, often driven by factors like lack of engagement, monotonous work environments, abundant job opportunities, and limited career prospects, remains a persistent challenge globally. Studies underscore that effective management of attrition not only mitigates business risks but also enhances organizational stability and productivity. Research efforts, such as those employing machine learning and statistical analyses, aim to predict attrition patterns and identify vulnerable employee groups. Various studies highlight the importance of understanding voluntary turnover dynamics and its implications for organizational strategies, including the deployment of early warning systems (EWS) to pre-emptively manage attrition risks. By leveraging data-driven insights and predictive models, organizations can proactively address attrition challenges and optimize workforce retention strategies. **(Mhatre, 2020)**

This work explores the critical issue of employee turnover in HR analytics, emphasizing its significant impact on organizational stability and costs. Employee turnover, defined as the departure of employees due to resignation, retirement, or other factors, poses substantial challenges to businesses, particularly in sectors with large-scale recruitment needs. The study leverages machine learning techniques such as logistic regression, decision trees, and random forests to predict and understand attrition patterns using data from a comprehensive HR dataset available on Kaggle. Key determinants of turnover include job position, overtime, and work level, highlighting the multifaceted nature of attrition drivers. The research underscores the importance of retaining skilled personnel through effective job satisfaction strategies, emphasizing the cost-effectiveness of retaining existing

employees over recruiting new ones. Results indicate that the random forest model outperforms other algorithms, achieving an accuracy of 90.20%, showcasing its efficacy in predicting turnover. The study concludes by suggesting future directions in employing deep learning on larger datasets to enhance predictive capabilities in workforce analytics. **(Chakraborty, 2021)**

This work examines employee attrition using IBM Employee Attrition data, focusing on identifying key drivers and predicting turnover using machine learning techniques. Various studies have investigated similar themes, often employing diverse methodologies. Alao et al. (2021) utilized decision trees and rule sets to predict attrition cases, while Alduayj et al. (2020) applied SVM and random forest for prediction, highlighting SVM's robust performance. Frye et al. (2019) employed PCA and classification methods like KNN and Random Forest, emphasizing logistic regression's predictive accuracy. Yadav et al. (2018) and Srivastava et al. (2017) developed frameworks using classification techniques to predict churn, underscoring the importance of behavioural attributes. El-Rayes et al. (2016) focused on predictive analytics for voluntary attrition. Setiawan et al. (2015) identified critical variables influencing attrition. This study contributes by employing Random Forest for feature selection and K-means Clustering for classification, revealing insights into demographic and job-related factors influencing attrition. Logistic regression further explores significant predictors such as gender, job role, and work-related factors. Overall, these findings underscore the multifaceted nature of employee attrition, urging organizations to tailor retention strategies based on nuanced predictive models and behavioural insights. **(Yang, 2020)**

This work explores employee attrition prediction using machine learning, focusing on the IBM HR Analytics Employee Attrition and Performance dataset. The study identifies significant factors influencing employee turnover, employing tree-based ensemble models. Previous research indicates a variety of algorithms applied to similar datasets, highlighting SVM, Decision Trees, Random Forest, and Gradient Boosting as effective tools. For instance, SVM and ensemble techniques like Random Forest exhibit accuracies up to 88.43%. The study emphasizes data pre-processing's role in enhancing model performance, including feature selection and scaling. Visualization tools such as KDE plots and correlation matrices aid in understanding data distributions and relationships. Model comparison reveals Random Forest achieving 86% accuracy, while Gradient Boosting outperforms with 95%. Overall, this research underscores the efficacy of ensemble methods in predicting and mitigating employee attrition, offering valuable insights for HR strategies and organizational planning. **(Mehta, 2021)**

This work explores the application of ensemble models for predicting employee attrition using IBM HR Analytics data. It compares single models like Logistic Regression, Random Forest, and Boost with ensemble methods combining these models. The study demonstrates that a stacking-based ensemble model, combining Random Forest and Artificial Neural Network as base learners with Logistic Regression as a meta-learner, achieves superior predictive performance. Key findings highlight the ensemble model's robustness in accuracy, precision, recall, F1-score, and AUC metrics, outperforming individual models. Feature importance analysis identifies variables such as Environmental Satisfaction, Relationship Satisfaction, Overtime, Performance Rating, and Monthly Income as significant contributors to attrition prediction. This underscores the importance of psychological and job-related factors in attrition modelling. Comparison with prior studies using the same dataset reveals the ensemble approach's efficacy in overcoming single model limitations, thereby enhancing predictive accuracy and informing strategic HR management. Future research directions include exploring additional factors influencing attrition and integrating qualitative analysis to enrich model interpretability and applicability in organizational settings. **(Chung, 2023)**

This work explores the prediction of employee attrition using deep learning techniques applied to the IBM HR Analytics dataset, which contains 35 features across 1470 employee records. Employee attrition, driven by factors such as work environment and job satisfaction, poses significant challenges

to organizational stability and productivity. Previous studies have employed various machine learning models, including random forests, SVMs, and deep neural networks, to predict attrition based on employee data analysis. For instance, Alduayj and Rajpoot utilized synthetic sampling methods to address data imbalance, achieving notable accuracy improvements. Usha and Balaji, meanwhile, emphasized the importance of pre-processing steps in enhancing model performance. In contrast, this study focuses on deep learning models, leveraging their ability to capture complex relationships within data through multiple hidden layers. Pre-processing steps involve data cleaning, categorical encoding, and feature scaling to optimize model performance. The evaluation utilizes both train-test splits and k-fold cross-validation to ensure robustness and generalizability. Results indicate significant accuracy gains, particularly with synthetic balanced datasets, highlighting the efficacy of deep learning in mitigating attrition risks. Overall, this work contributes to advancing predictive capabilities in HR analytics, aiding organizations in pre-emptively addressing attrition through data-driven insights. **(Al-Darraji, et al., 2021)**

This work explores the critical issue of employee attrition in software companies, emphasizing the significant impact on organizational stability and growth when skilled employees depart. Utilizing the IBM HR Analytics Employee Attrition and Performance dataset, the study underscores the necessity for predictive analytics to pre-emptively identify and mitigate attrition factors. Initial efforts employ sampling techniques—random oversampling, random under sampling, and SMOTE—to address data imbalance, crucial for accurate model performance. Various classification algorithms including logistic regression, K-nearest neighbour, decision tree, random forest, and AdaBoost are applied to analyse and compare their effectiveness in predicting attrition risk. Prior studies have similarly utilized machine learning models to predict attrition, highlighting key variables such as job involvement, years with the current manager, and overtime as significant predictors. The findings underscore the importance of tailored HR strategies informed by predictive analytics to retain high-performing talent and sustain organizational growth. **(Vinoodhini, 2022)**

2.1 Discussion

The programming language chosen primarily in current project is Python. It can analyse staff departures in no small measure thanks to its significant advantages in data science and machine learning. Python is known for its readability and simplicity, which are crucial for complex data analysis procedures. With a well-developed ecosystem of libraries and frameworks, it covers a wide variety of tasks from handling information through machine learning. This makes the language attractive working tool to data scientists and analysts. Besides, the language's popularity is also supported by its strong community and continuing development, so it remains a cutting-edge tool in the data science arsenal.

Data — IBM HR Analytics Employee Attrition & Performance This dataset is rich, with many features to represent employee demographic information (age, etc), job characteristics and performance metrics as such. It is an abundant dataset and therefore it can be analysed in depth to ascertain all the factors which lead employees to leave, thus perfecting a broad approach of information required for detecting patterns and trends of employee turnover.
Data pre-processing is the most important step in any data analysis project to get a clean and accurate dataset. After that, we can proceed with model building. This project has also performed some pre-processing techniques to treat missing values, outliers, and categorical variables. Other common reasons that may introduce missing values in a dataset are data entry errors or non-responses during surveys. This was addressed via mean and median imputation techniques. The overall mean, as an estimation of the continuous value in a numerical feature with almost normal distribution, was imputed since we assume missing values would be randomly located close to or equal to means. For certain features with skewed distributions, median imputation was used instead. The median is less

sensitive to extreme values, making it a robust choice for skewed data. This method ensured that the data was not corrupted due to large amounts of missing values.

An outlier is a data point that varies significantly from other observations, which can skew statistical analyses and model performance. Outliers were identified by the Z-score method. This approach will essentially determine how many standard deviations a data point is from the mean, and anything that falls outside of some threshold Z-score is marked as an outlier. This helped in not getting skewed-up results because of those extreme values, and to get output according to the rest.
Categorical variables — discrete categories (i.e., no numerical encoding) => one-hot ecotype. One-hot encoding consists of transforming categorical variables into a series of binary categories with only one present in each. And then, of course, we have to change their original form because, for most machine learning models, only floats are allowed. Conversion from values like "Sales," "HR," and "IT" (Department) to three binary columns representing the presence or absence of each department—for example. This method helps in the proper representation of categorical variables in analysis.

Exploratory Data Analysis (EDA) is necessary to discover the patterns and relationships of data. Create a summary and visualize the data to get useful insights. After EDA, data wrangling techniques are used to improve the comprehension of data — for instance, combining sources, picking important attributes, or creating new features. EDA is incomplete without tools for Data visualization using Matplotlib and Seaborn. We also need to understand the attrition rate by age or income type, which can be done using histograms because they show central tendencies and distributions of variables such as age or income. Such values could betray slight differences into joined box plots (e.g., income by job role) that are going unnoticed in central tendencies. Bar plots of attrition rates by group (such as department or job role) can quickly show trends in which groups are most affected.

Statistical analysis was performed to check the significance of different factors responsible for employee attrition apart from EDA. We made use of two important statistical methods — ANOVA (Analysis Of Variance) and the Chi-Square test.

1. ANOVA (Analysis of Variance): ANOVA is a popular statistical method developed by R. A. Fisher himself and used to compare the means of numerical features across different groups, i.e., stayed vs. left employees, etc. ANOVA computes sums of squares between groups and within the group, which helps in identifying if there is any difference in salary or age (features). This is used to identify which numerical variables could affect employee attrition. For example, if ANOVA detects that there is a significant difference between employees who stayed and those who quit in terms of salary, we can infer how important salary might be to retention.
2. The Chi-Square test, in contrast, is used to evaluate the relationship between two categorical variables. It tests whether the distribution of one categorical variable (for example, levels of job satisfaction) is independent from another variable (attrition status). The Chi-Square test indicates, through observed versus expected frequencies, whether or not particular categories exhibit high or low turnover. For instance, if there is a high degree of relationship between low job satisfaction and high attrition rates, we will consider job satisfaction as having certainly contributed to increasing the employment turnover rate. ANOVA and Chi-Square are equally important to comprehend the cause of employee attrition and intervene in a targeted manner.

EDA and statistical analysis together gave a detailed insight into variables affecting attrition for the employees. Identifying Key Variables and Relationships Using Numerical and Categorical features — We analysed numerical and categorical attributes to identify the most important variables contributing to turnover. This information is invaluable when it comes to devising strategies regarding employee reductions or increasing satisfaction and overall organization performance.

Both studies highlight the need for a mix of data pre-processing, exploratory analysis, and statistical methods to examine employee attrition effectively. Python's libraries and tools played a key role in conducting this deeper analysis that can help identify important factors/patterns for the business, which they can use to make strategic decisions to reduce employee turnover.

2.2 Comparison With Existing Literature

Employee attrition has been a rapidly evolving field of study with the advent and democratization of machine learning (ML) in general, but more specifically within HR analytics. A variety of methodologies and algorithms are used to make such accurate predictions; more on these in the literature review, along with predefined HR datasets that allow researchers to help business owners prepare for employee churn.

The literature review mostly discussed different ML algorithms used for attrition prediction, with commonly targeted models being Random Forest, Support Vector Machines, Logistic Regression (LR), and Naïve Bayes. Across studies, ensemble models (most consistently Random Forests) have performed well in more complex problems and provided high levels of predictive accuracy. In particular, the work by Chaitanya and Srivastava (2020) showed that RF dominates many other classifiers because it performs well in learning nonlinear relationships and regularizes gridding problems, a view also shared by works like (**Fallucchi,** 2020) and **(Krishna, 2022)**.Likewise, ensemble methods like AdaBoost and Gradient Boosting have also shown good performances, as reported in the studies of **(Ozdemir, 2020)** and **(Mehta, 2021).**

Nevertheless, there are quite a few studies that have found certain instances when other models, like SVMs, tend to do better. In the attrition prediction context, for instance, Sharma and Goyal (2018) cited SVM as having shown good accuracy due to their capability in handling high-dimensional data with non-linear boundary conditions. This finding is similar to that reported by **(Mansor, et al., 2021)**,who found SVMs to be the best model among others like Decision Trees and Artificial Neural Networks (ANN), but only after rigorous parameter tuning. The application of these methods on different datasets suggests that while ensemble methods are inherently stable, the choice of the best algorithm is highly dependent upon dimensionality and feature selection, significantly influenced by the specific context in which they are applied

In addition to predictive accuracy, another critical topic explored in the literature is which factors matter when it comes to attrition. Yet surveys keep showing that the single most important reason employees leave a job is not high stress or long hours; it's lack of engagement. The work of Hausknecht et al. (2009) and Hom et al. (2017) has helped establish this paradigm of central importance for understanding employee behaviour. The present study also builds upon these variables by adding them to the feature set of the model used for training, thereby underlining their significance in accordance with previous works.

Moreover, the literature highlights that well-processed data is essential to increase model efficiency, particularly in feature reduction and handling missing values. The methods used to handle missing values and outliers, like mean and median imputation for missing data and the Z-score method for outlier detection, are all well-established in the literature. For example, a study by **(Al-Darraji, et al., 2021).** emphasizes the importance of pre-processing in enhancing model resilience and generalizability. One-hot encoding of categorical variables, as done in this project, is also a common practice to ensure ML models that expect numerical input can process the data effectively, as discussed by **(Najafi-Zangeneh, et al., 2021)**

Given this, the rise of new neural networks and ensemble learning methods is becoming prominent in HR analytics, mirroring trends seen in broader AI research. Studies like those by **(Chung, 2023)** and Alao et al.(2021) reflect the growing attention to these approaches and also noted the increasing presence of deep learning, which not only excels at capturing complex data patterns but also offers improved predictive performance across various applications.

In contrast, the current study focuses on ANOVA and Chi-Square, while also pointing out how well ML algorithms complement them. This hybrid approach of combining traditional statistical methods and machine learning helps to identify significant factors using ANOVA or Chi-Square tests while predicting attrition rates more accurately. It also captures interaction effects that might have been missed by normal predictive models, thereby providing a robust solution for understanding and managing employee attrition.

The literature also has recurring concerns related to ethical considerations and data quality. There are ethical questions about fairness and transparency when using ML in HR, and scholars like Cappelli (2019) have pointed out problems with bias in prediction. To tackle this issue, the project emphasizes data quality and ethical concerns in the model development process. The literature suggests that future research should focus on these areas, specifically on developing models that are not only accurate but also fair and interpretable.

In summary, the use of ANOVA and Chi-Square in this study correlates with typical HR analytics techniques. Like the broader study, each of these models is good for distinguishing group differences or understanding relationships between categorical variables and attrition, with detailed attention toward the most salient factors that lead to an exit (key attrition factors) combined with rigorous data pre-processing. However, the literature also shows that there is a lot of potential left unexplored; particularly, for using more advanced methods such as deep learning in HR analytics and understanding ethical dilemmas surrounding ML in HR. Future research should work towards improving these models, making them more interpretable, and applying the criteria to be not only accurate but fair in how they are implemented so that their potential influence on organizational decision-making is maximized.

Chapter 3. Methodology

The methodology for this project aims at doing an EDA of the IBM HR Analytics Employee Attrition & Performance dataset using Python. The purpose is to reveal the reason behind what number of employees are leaving the organization and which factors are effective on employee performance. It comprises of Data Pre-processing (to clean and process the data), EDA to find patterns and trends, Statistical analysis techniques to check for feature significance etc. Secondary data sample – greater efficiency, less costs. The main focus is on ethical considerations, with data anonymization and confidentiality preserved throughout the analysis. The results should be used to guide our strategies for increasing the persistence and performance of employees.

3.1 Research Design and Approach

The objective of this project is to use an exploratory research design for analysing the factors affecting employee attrition and performance. The study uses the IBM HR Analytics Employee Attrition & Performance dataset and applies different data analysis techniques to detect patterns and trends. By investigating, one can truly expand all about the dataset and locate advantage areas where in fact the organization should focus for optimizing employee retention & performance management. Such a design facilitates an in-depth look at the factors driving employee behaviour and performance to provide valuable insights for creating focused solutions. This study aims to stimulate debate and

provide a practically useful foundation for managing humans in the firm, which can decrease attrition rates while improving performance.

3.2 Data Collection Methods

The data for this project was sourced from Kaggle, an online platform that provides extensive datasets. The analysis was on an employee attrition and performance dataset obtained from Kaggle, which consists of 35 features (both demographic such as age, etc., job-related like salary component or level of education attained by a person, and company-specific features). This publicly available, anonymized dataset adheres to ethical research standards as the data has been used in many data science research, establishing a broad range of reliability. Kaggle was selected due to its credibility as a source for good quality, well-documented datasets, which could be of use in determining reasons why an employee might leave and subsequently inform organizational prevention strategies.

3.3 Sample Selection and Participants

The experiment is based on the data availability in this organization from various departments. We had used IBM HR Analytics Employee Attrition & Performance dataset, which includes information on employee demographics e.g. age, department, and job role in the company. It also includes work environment ratings (evaluation), satisfaction level with their current roles compared to previous roles, etc. Being an anonymized data set, they respect employee privacy, and there are no distinctive participants chosen. This sample size is thought to be large enough to allow for some interesting findings about which factors really drive attrition and performance. The sample is representative of the overall employee population, and therefore gains in external validity, allowing some confidence in drawing results with the potential to help companies improve their employees' retention rates and thus organization performance.

3.4 Data Analysis Techniques

3.4.1 Data Pre-Processing

The most crucial stage in any data analysis is data pre-processing. In order to facilitate the execution of machine learning algorithms on the data, it must first be extracted from its original location and prepared for analysis through cleaning, transformation, and organization. These are the steps involved in pre-processing data:

Loading Dataset: The analysis environment gets the dataset from Kaggle, which provides a structured base for study.

Insights of Dataset: It is the first understanding of the dimensions of the dataset, columns in each row, and numbers. It helps you get familiar with how your dataset looks like and possible study fields.

Data Wrangling: This process includes handling missing values, removing duplicates, correcting errors, and dealing with inconsistent data formats.

- **Handling Missing Values**: Replace missing data with appropriate values (mean, median, mode) or drop the rows/columns containing missing data.
- **Dealing with Outliers**: Detect and handle outliers that could skew the analysis.
- **Feature Engineering**: Creating new features from existing data to improve the model's performance.
- **Encoding Categorical Variables**: Convert categorical data into numerical formats using techniques like one-hot encoding or label encoding.

3.4.2 Exploratory Data Analysis

EDA (Exploratory Data Analysis) is the very first step in our analysis process to understand and get a sense of data through descriptive statistics and graphs. This gives us a peek of what we can expect in the data by seeing and decoding different distributions using tools like distribution plots, histograms, or box plots for numerical variables. You can also visualize bar or pie charts for categories. A brief snapshot of the data can be gleaned quickly by looking at these visualizations and may save us a ton of time in spotting patterns, trends, or possible outliers.

Employee Attrition Trend Visualization: Displays the attrition rate of your organization at a macro level and compares it with industry benchmarks to understand any potential root causes behind employee churn and its impact.

Employee Attrition by Gender Analysis: This area analyses employee attrition in terms of male and female employees, helping to understand if there is a problem targeted at either gender.

Analysing Employee Attrition by Age: Find out how attrition rates differ among various age groups and get insights into the retention problems younger employees face.

Evaluating Department-Wise Employee Attrition: Studies employee attrition according to departments, providing insights into potential department issues like high turnover in Sales and HR.

Analysis of Attrition by Distance From Home: Investigates whether the distance an employee commutes to work is related to their attrition, highlighting non-work factors in staff turnover.

Analysing Employee Attrition by Education: Analyses the effect of an employee's educational level on their likelihood to leave, possibly showing that lower educational levels are more likely to experience attrition.

Analysing Employee Attrition by Job Roles: The range of job roles is analyzed to focus on positions with high turnover, leading to more targeted retention strategies.

Analysing Attrition by Job Level: This provides insight into how attrition rates vary across different roles, concluding that junior positions have slightly higher turnover and exploring the probable reasons.

Analysing Employee Attrition by Job Satisfaction: Some satisfied workers may still quit, and dedicating a section to show the relationship between this variable reinforces the need to look beyond surface-level categories.

Analysing Employee Attrition by Marital Status: Identifies the relationship of marital status on attrition, with higher turnover noted among divorced employees.

Attrition Analysis with Monthly Income: This visualization indicates that employees with lower monthly income are more likely to leave, emphasizing the need for better compensation.

Analysing Employee Attrition by Percentage Salary Hike: This analysis shows that employees who have received a lower percentage salary hike are more likely to leave.

Employee Attrition in Years by Current Role: Studies how the number of years an employee has spent in their current role impacts turnover, with high attrition recorded for those with 2-5 years in their role.

Analysing Employee Attrition by Years Since Last Promotion: Examines the impact of time since the last promotion on attrition, noting that employees who have experienced longer gaps between promotions are more likely to quit.

Analysing Employee Attrition by Years at Company: Looks at the overall years spent working for a company to determine which employees are more likely to leave.

3.4.3 Statistical Analysis

- **ANOVA Test for Numerical Features**: Identifies the impact of numerical features on employee attrition by assessing the significance of each feature's variance in relation to the categorical target.

  - **Statistical Values**:
    - **F-Score**: Measures the strength of association between numerical features and attrition.
    - **P-Value**: Determines the statistical significance of each feature's impact on attrition.

Visualizing F-Score of ANOVA Test: Graphically represents the strength of association for each numerical feature based on F-scores, helping identify the most influential factors in employee attrition.

Comparing F-Score and P-Value of ANOVA Test: Compares the F-scores and p-values of numerical features to pinpoint those with the most significant impact on attrition, guiding targeted retention strategies.

Chi-Square Test for Categorical Features: Assesses the association between categorical features (like job role, department) and employee attrition, identifying key factors influencing turnover.

Visualizing Chi-Square Statistic Values: Bar plots illustrate the strength of association for each categorical feature, highlighting those most strongly linked to employee attrition.

Comparing Chi-Square Statistic and P-Value: Evaluates the statistical significance of categorical features by comparing Chi-Square statistics and p-values, helping prioritize factors for targeted retention efforts.

3.5 Significance of Employee attrition and Other Columns

In this section, we look at the individual variables that affect employee attrition to a massive extent in the organization. Roles of each variable influencing employee retention or turnover are explained as follows:

Employee Attrition: It tells us whether an employee has left the company (Yes) or stayed. Knowledge about the determinants of an independent variable helps to implement specific corrective measures to reduce attrition rates.
Gender: Analysis of the sex composition seeks to identify whether there are important dissimilarities between men and women in attrition. While a lesser predictor most of the time, gender evaluation may reveal if some sex-connected factors are leading to above-average attrition in one group.

Age: Experience is important, but so is the relationship with age (which often relates to career stage and job satisfaction). Younger employees might leave to explore careers, while older employees might be approaching retirement.

Department: An observation of which division has higher employee turnover. For example, Research and Development might experience higher turnover due to skill demand or job stress.

Distance from Home: This variable shows how far an employee lives from the workplace. High commute times can lead to dissatisfaction and potential attrition. The analysis reveals if proximity is a significant factor in staying or leaving.

Education: Educational background determines career progression and satisfaction. Advanced roles typically require higher education, and lower levels may be associated with higher attrition.

Job Role: Different job roles in the organization have varying attrition rates based on factors like job satisfaction, career opportunities, and work-life balance. High-stress roles or those with limited growth may see higher turnover.

Job Level: This reflects an employee's position in the hierarchy. Higher-level jobs often have lower turnover due to better pay, benefits, and satisfaction.

Job Satisfaction: Job satisfaction is critical—it directly influences whether an employee stays or leaves. Research shows that younger employees often have lower satisfaction, leading to higher turnover.
Marital Status: Marital status may affect job satisfaction and turnover. Married employees often prioritize work-life balance, which can influence their decision to stay.

Monthly Income: Income is a strong attrition predictor. Employees are more likely to leave if they believe they can find better-paying jobs elsewhere. Higher income generally correlates with lower attrition.

Percent Salary Hike: Significant raises can boost morale and retention, while minimal increases may lead to dissatisfaction and higher turnover.

Years in Current Role: The length of time in the current role may indicate job satisfaction and growth. Employees in the same position for too long without progression may feel stagnant, increasing attrition.

Years Since Last Promotion: This variable shows the time since the last promotion. Long-serving employees may feel undervalued without promotions, leading them to leave, especially if they see limited career advancement.

Years at Company: This shows an employee's tenure. Short and long tenures impact attrition differently, with motivations varying between early-career employees and those nearing retirement.
A deeper analysis of these variables helps the organization design specific strategies to ensure higher employee satisfaction and retention, reducing attrition rates and improving productivity.

3.6 Ethical Considerations

This research concerns ethics and the right to privacy or data security, hence it requires ethical considerations. It is an anonymized dataset; all references to the identity of employees have been removed. This study involves secondary data analysis using a publicly available dataset; therefore, it does not require informed consent.

Data security and privacy: This dataset has a robust privacy policy with detailed procedures for data security, illustrating that the research based on this dataset also respects ethical practices. This is very important in preserving the integrity of the research and ensuring that the results are ethical. The

ethically sound method is to prevent any harm accruing from the research toward those whose data is being processed, following strict regulations for responsible big data use.

CHAPTER 4. Results

4.1 Presentation of Findings

The purpose of this study was to grow a mission-driven organization, uncover them research questions and tell you the story by category that are responsible for employee's attrition. Using meticulous Exploratory Data Analysis (EDA) and statistical tests, the analysis flagged several meaningful attributes linked to increased attrition. But also key variables such as job role, monthly income, level of job satisfaction, and number of years at the company became evident elements to understand why employees leave.

For instance, the data showed that employees in certain job roles with lower satisfaction levels or those earning less were more prone to leaving the organization. Additionally, years spent at the company demonstrated a clear relationship with attrition, where employees in their early years or nearing retirement were more likely to exit. These patterns were visualized through various charts and graphs, including bar charts, histograms, and box plots, which made it easier to discern the differences between employees who stayed and those who left.

Moreover, the findings highlighted the importance of understanding both individual attributes, such as satisfaction and income, and organizational factors like job roles in predicting attrition. The visualizations provided a clear and comprehensive view of these relationships, offering actionable insights for organizational strategies aimed at reducing turnover.

In summary, the findings underscore the complex interplay between personal and job-related factors in employee attrition, suggesting that targeted interventions in these key areas could help mitigate turnover and retain valuable talent within the company.

4.2 Data Analysis and Interpretation

The analysis and interpretation phase of this study focused on validating the findings through rigorous statistical testing and data exploration. The data analysis process involved several steps, each aimed at uncovering the most relevant factors contributing to employee attrition.

I. **Statistical Testing**: Two main statistical tests were conducted: ANOVA for numerical variables and Chi-Square for categorical variables. The ANOVA test revealed significant differences in the means of numerical features like age, income, and years at the company between those who stayed and those who left. The Chi-Square test confirmed the association between categorical variables (like job role and department) and attrition rates.

II. **Interpretation**: The statistical results were interpreted in the context of the broader organizational environment. For instance, low satisfaction levels among certain job roles were identified as a major driver of attrition, suggesting the need for targeted interventions in these areas.

These findings underscore the importance of addressing both organizational and individual factors in any strategy aimed at reducing attrition.

4.2.1 Data Pre-Processing

The first step before analysing data is the pre-processing phase designed to assure that produced results are valid, reliable, and correct. The main steps taken were the following:

I. **Data Cleaning**: Missing values, duplicates, and outliers in the dataset; Where data was missing, it did not negatively bias the estimates and this information is either imputed or recorded for when used . The outliers were then singled out and it was determined whether they are legitimate data points or erroneous.

II. **Data Transformation**: One-hot encoding techniques to encode categorical data, whereas numerical data required normalization for further analysis. For instance, salary data was normalized to facilitate benchmarking between different segments of our workforce.

III. **Feature Selection**: To filter the most effective features, we did an extensive feature selection process to do away with unnecessary features in order to focus on independent variables that have a high impact on the dependent variable. Part of this process was to filter out features that were redundant or would distract the analysis—all with the goal of being both efficient and effective with our data.

IV. **Imbalanced Data**: Since attrition is an imbalanced target variable, techniques like oversampling or under sampling were incorporated to make sure models used in the study would be able to predict the same with a fair amount of accuracy.

4.2.2 Exploratory Data Analysis (EDA)

1.Visualising the Employee Attrition Rate:

The organization's attrition rate of 16.12% is quite alarming compared to industry benchmarks ranging from 4%-6%, according to Human Resources experts. Most likely, this high attrition rate is a result of deeper issues, such as an unhappy workforce or a lack of career development opportunities and life balance. This well-above-average rate is dangerous and can result in higher recruitment and training costs, loss of long-term institutional knowledge, decreased employee morale, and operational disruptions. There are inherent risks that the organization must control by leading more emphasis on employee engagement, work-life balance activities, and so forth to keep good compensation behind. These steps will additionally include conducting exit interviews, allowing the company to gather critical feedback on why employees are leaving and build up weaker points. The above will enable the organization to more adequately address key issues, resulting in less attrition and greater workforce stability, with overall better performance over time.

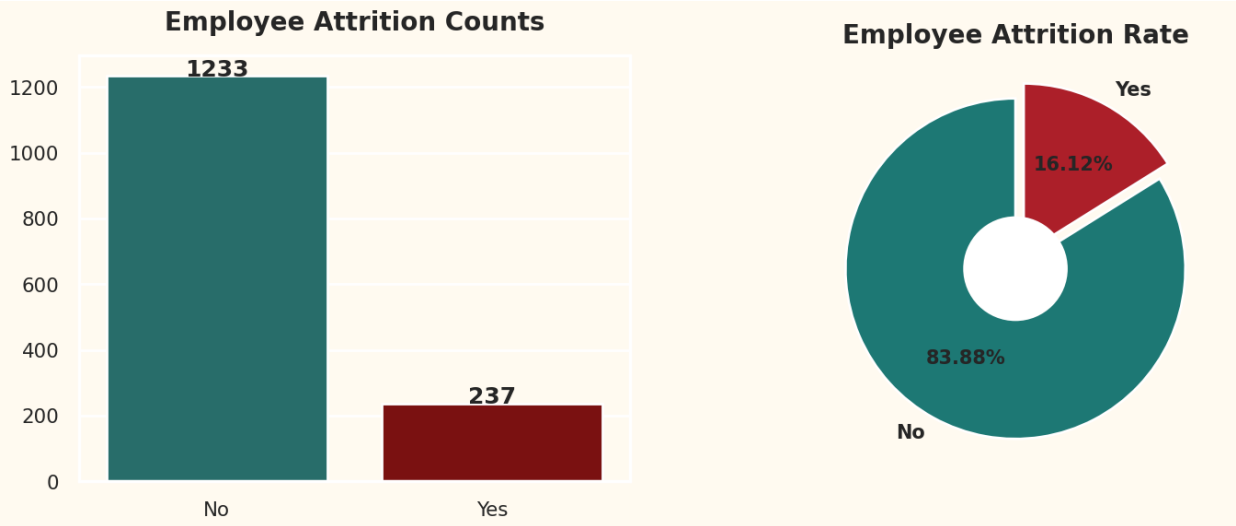


Figure 1. Visualising the Employee Attrition Rate

2.Analysing Employee Attrition by Gender:

The organization is somewhat male-dominated, with men making up just over 50% of the employees. Even though there are more men leaving the organization than women. This trend could hint at problems that may exist in the company for male employees. It is important to identify the reasons behind this male/female attrition in order to tackle these problems at their source, improving overall employee retention.

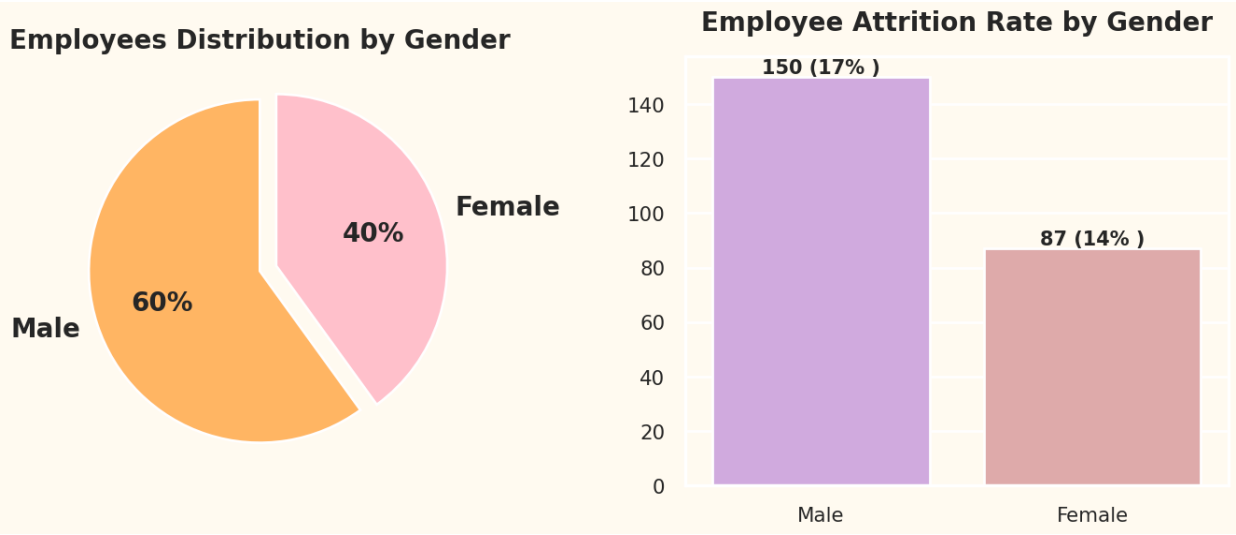


Figure 2.Analysing Employee Attrition by Gender

3.Analysing Employee Attrition by Age:

The company has an employee base where the majority are between 30–40 years old. There is an evident trend—attrition rates decrease with age. This means the younger workforce is more susceptible to leaving the organization compared to higher age groups. This is also confirmed by the boxplot chart, which shows that the median age of former employees is lower than those who remain. This indicates that younger staff members are either experiencing difficulties or have different professional requirements that result in their exit, highlighting the need for targeted retention strategies for this age group.

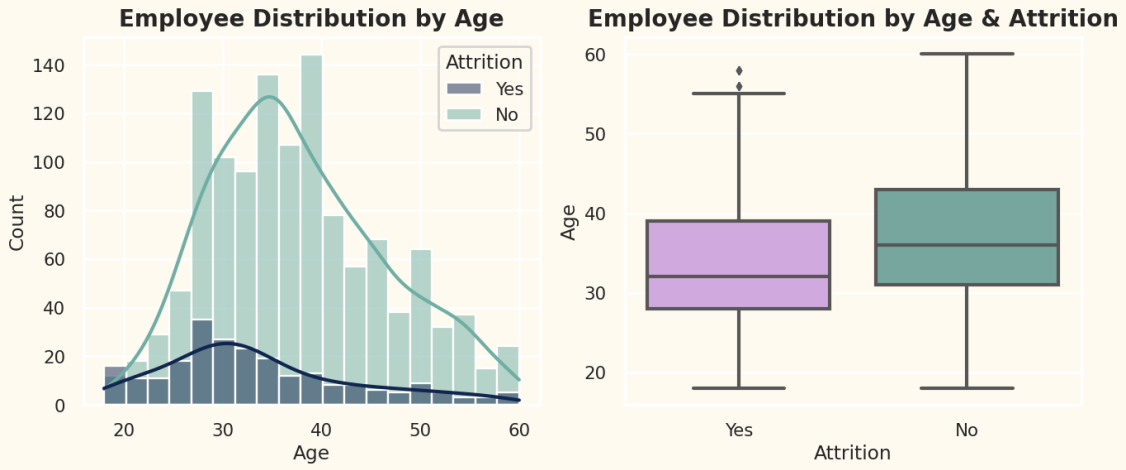


Figure 3.Analysing Employee Attrition by Age

4.Analysing Employee Attrition by Department:

The Research and Development (R&D) department has the largest number of employees in most organizations; however, despite this fact, it experiences the lowest attrition compared to other departments. Meanwhile, the Sales department retains a few employees yet experiences the highest attrition rate (or staff turnover), meaning that they have effective hiring practices. The Human Resources department has a worryingly high attrition rate too, considering their responsibility for the well-being and retention of employees themselves. These trends indicate that the R&D department has already normalized, but it may be beneficial to target specific issues related to high turnover in both Sales and HR departments.

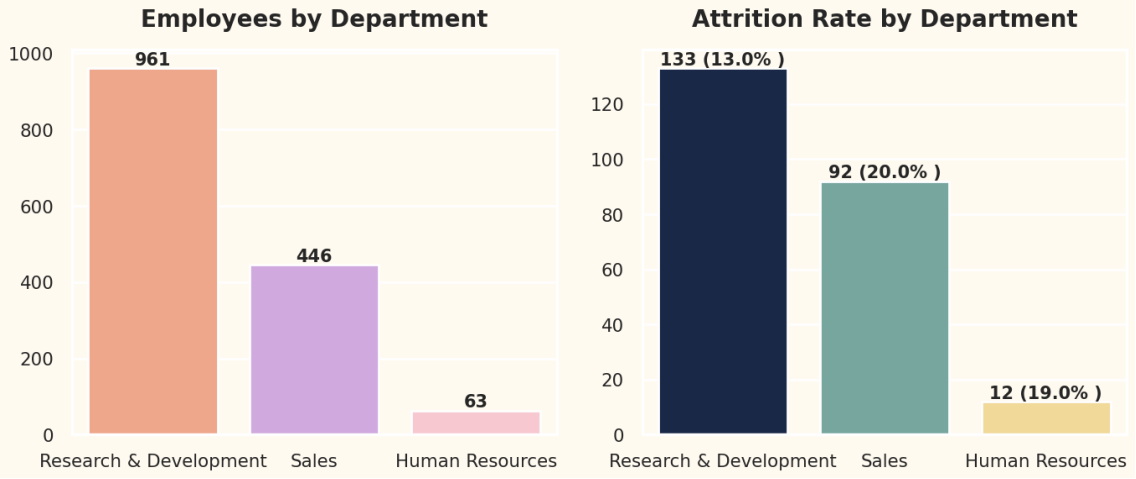


Figure 4.Analysing Employee Attrition by Department

5.Analysing Employee Attrition by Distance From Home:

Its workforce is comprised of people both near and far from the office. It is to be noted that "Distance From Home" does not show any clear trend with respect to the attrition rate. Nevertheless, residents in closer proximity to the entity turnover more frequently than their counterparts who live a good distance from it. This implies that it is not the vicinity to the office which determines retention, but there could be other reasons explaining why employees staying nearby are favouring a higher attrition rate.

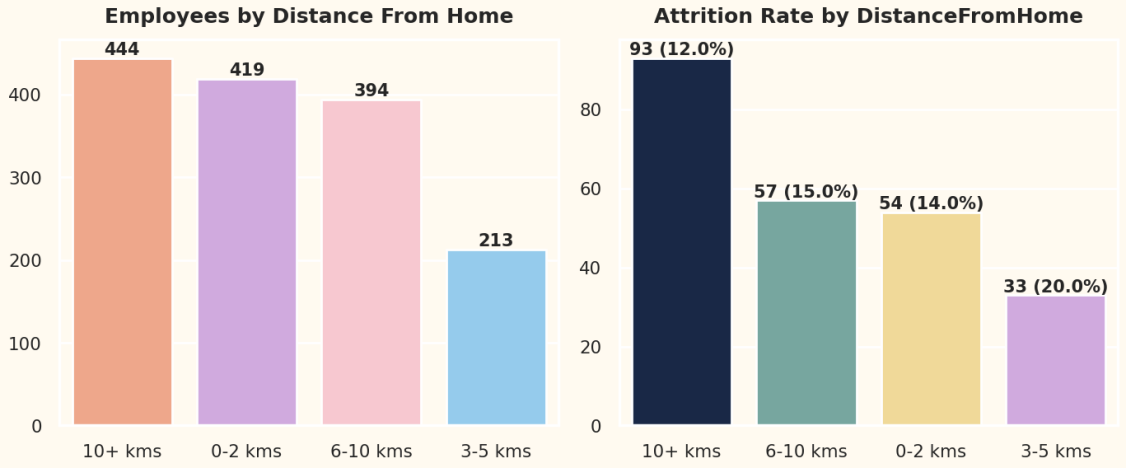


Figure 5.Analysing Employee Attrition by Distance From Home

6.Analyzing Employee Attrition by Education:

Most employees in the organization are at the Bachelor or Master degree level, with only a few holding a Doctorate. There is a distinct pattern where, as the level of education increases, attrition rates go down. Factors such as tenure, education level (Master's and Doctorate), which can reflect job satisfaction, career opportunities, or a person-organization fit, also have an influence. On the other hand, workers with less advanced education levels might be inclined to leave, indicating that tailored retention initiatives are needed for them.

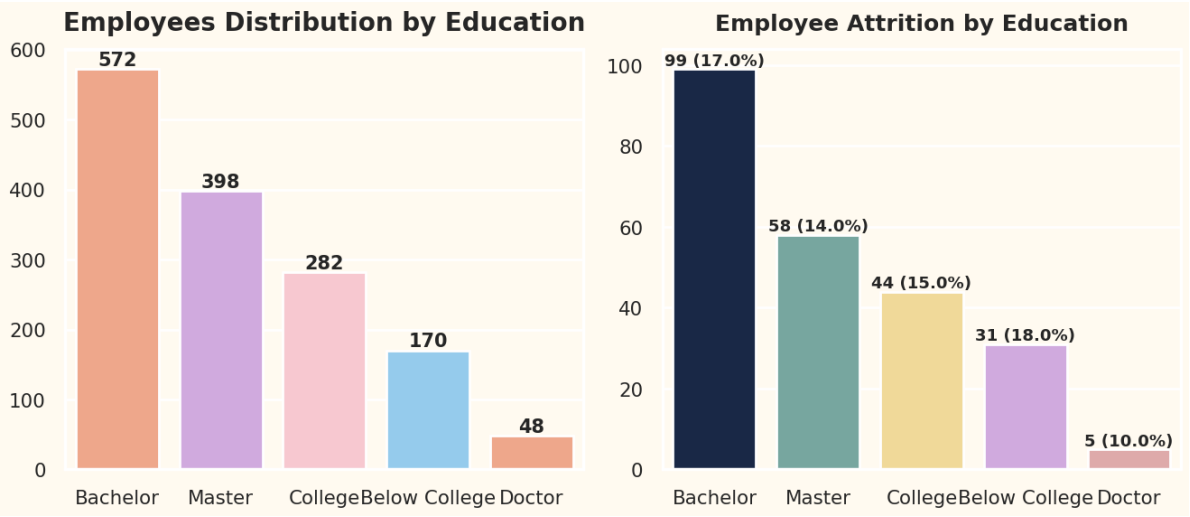


Figure 6.Analyzing Employee Attrition by Education

7.Analysing Employee Attrition by Job Roles:

Most employees were hired in the organization as Sales Executives, Research Scientists, and Lab Technicians. Research Directors, Sales Executives, and Research Scientists have the highest attrition rates. This tells us that these roles are crucial to the organization, but they suffer high levels of attrition—particularly at senior and front-line levels, like Sales Executives. Higher attrition in this area could imply that these roles are more stressful, require more effort, or possibly do not deliver on career expectations, highlighting the need to establish initiatives catered towards retaining talent in these critical function areas.

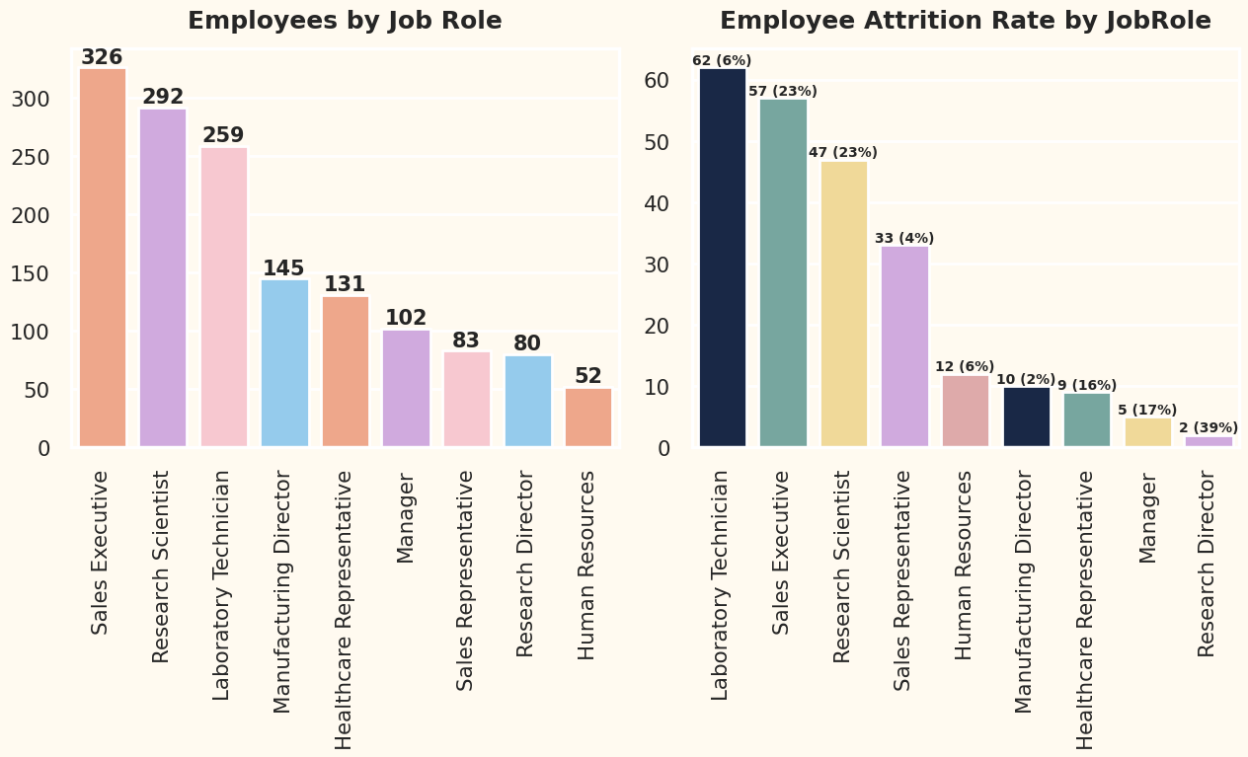


Figure 7.Analysing Employee Attrition by Job Roles

8.Analysing Employee Attrition by Job Level:

Most of the employees are at either Entry Level or Junior Level positions. Most importantly, the attrition is highest at the Entry Level by a big margin, whereas you can see that attrition declines as employees move to higher levels within the company. Relocating early job positions corresponds with higher turnover rates, suggesting that entry-level employees are hitting roadblocks, whether in the form of limited growth paths, dissatisfaction with their new jobs, or insufficient support from the employer. Solving these problems through better onboarding, career development programs, and mentoring could lead to lower attrition at entry levels and higher organic turnover as employees grow within the organization.

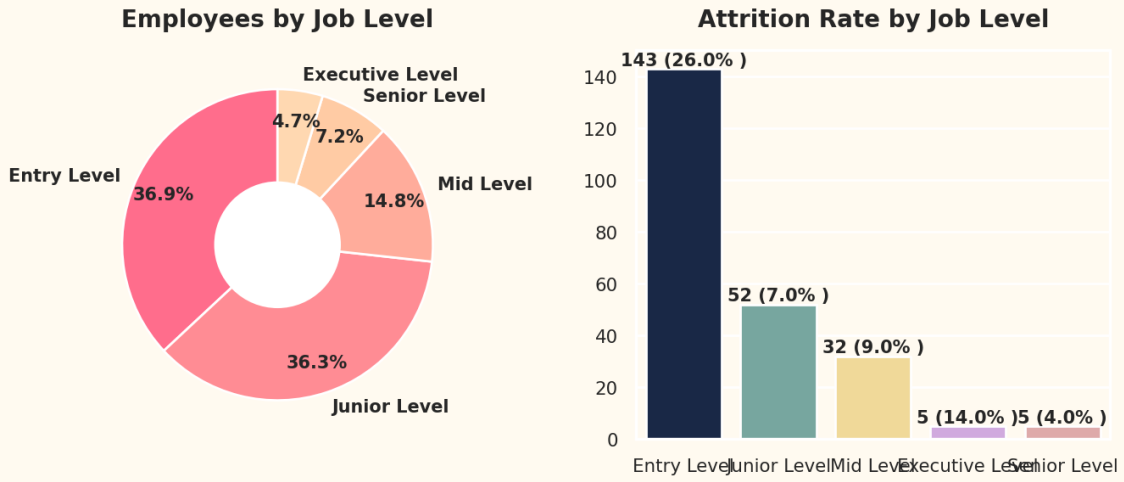


Figure 8.Analysing Employee Attrition by Job Level

9.Analysing Employee Attrition by Job Satisfaction:

As described by most employees in the organization, satisfaction with work is high or very high. Even more interesting is that despite the fact everyone—whether "satisfied" or "not satisfied"—mostly enjoys their work, all categories are experiencing high levels of attrition. While this does suggest that retention is tied to job satisfaction, it may not be the only factor forcing workers out of these jobs, and a deeper analysis into all causes of such widespread turnover should be performed.

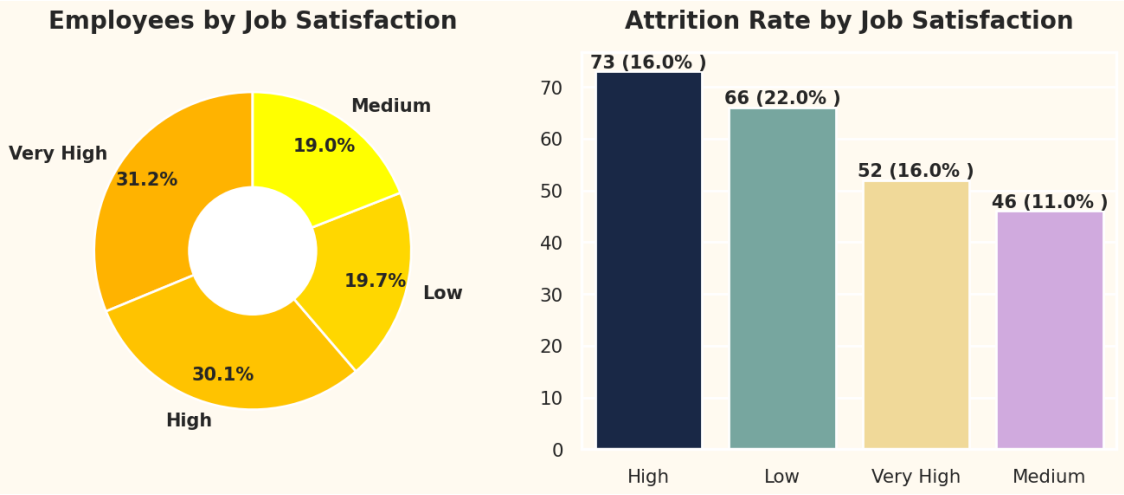


Figure 9.Analysing Employee Attrition by Job Satisfaction

10.Analysing Employee Attrition by Marital Status:

Nearly everyone in the organization is married. That said, divorced employees are significantly more likely to leave than workers who are married, as well as their single counterparts. Single employees, on the other hand, do not suffer high levels of attrition. This discrepancy could indicate that employees going through a divorce might have specific challenges or stressors that lead to higher turnover. Tackling these particular issues and offering the right level of support may help reduce attrition among this demographic, ultimately increasing the overall employee retention rate.

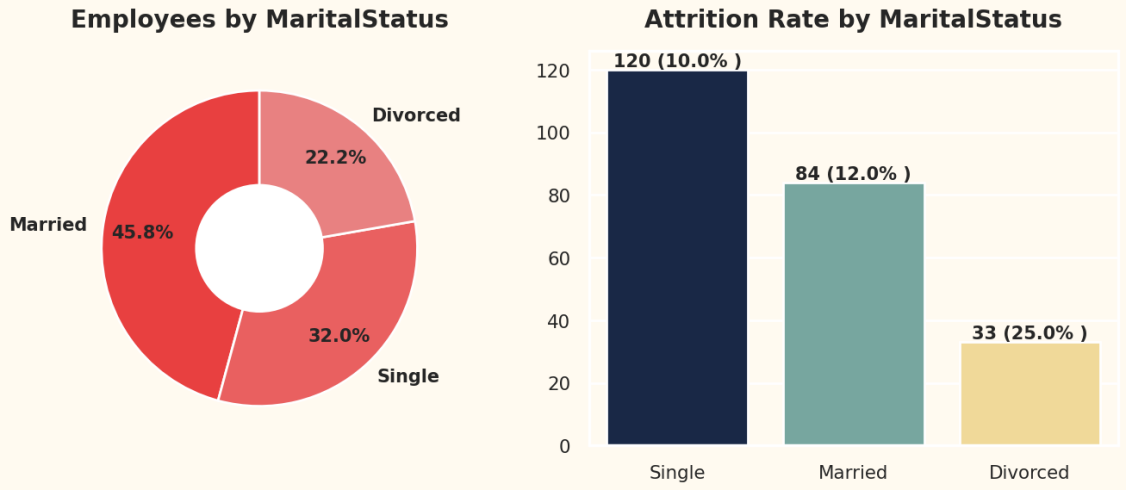


Figure 10.Analysing Employee Attrition by Marital Status

11.Analysing Employee Attrition by Monthly Income.

Most of the employees in that organization are earning below 10,000 per month. At this stage, you can notice that the average monthly income of leaving employees is usually less than those staying. The plot also elucidates a distinct trend — attrition tends to drop as monthly income increases. This may mean that higher-paid positions lead to lower turnover and offer a glimmer of hope for employers upping their compensation game as an employee retention strategy. Closing the wage gap and making sure that salaries are competitive with industry averages can be key to decreasing attrition rates and some small gratification on any job.

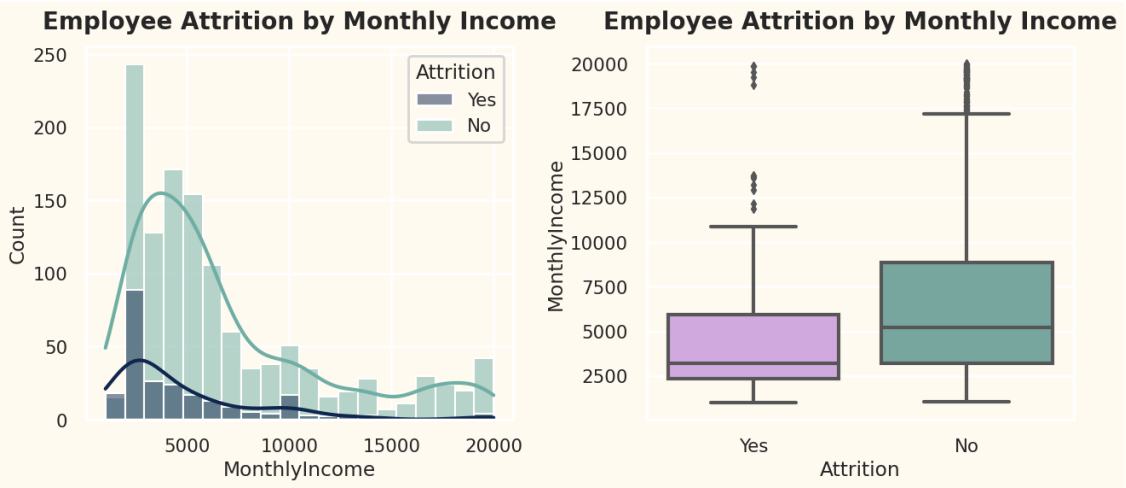


Figure 11.Analysing Employee Attrition by Monthly Income

12.Analysing Employee Attrition by Percentage Salary Hike.

Only a small number of professionals in the organization get such percentage increments in salary. But an interesting trend that emerges as you move over the % increase in salary—attrition rate is when pay increases faster, your staff are less likely to leave. This indicates that the employees who get more salary hikes in percentage also have a low intention to leave the organization. Increased salary hikes that are in sync with employee performance and market standards could serve as a good measure to heighten their retention while simultaneously curbing attrition.

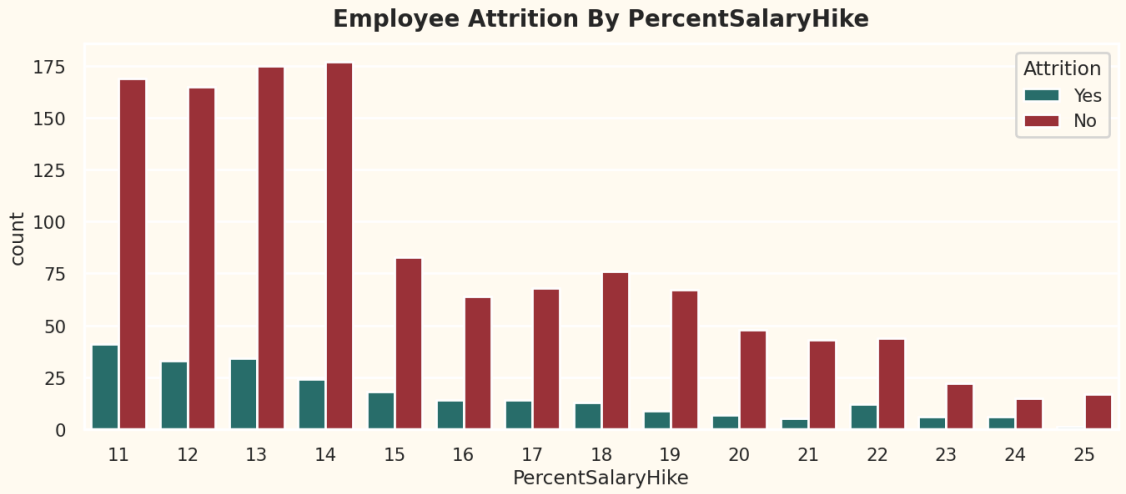


Figure 12.Analysing Employee Attrition by Percentage Salary Hike

13.Analysing Employee Attrition by Years In Current Role.

The majority of the organization's employees have only held their role for 2 to 10 years, with very few having been in it less than a year or more than ten! However, 2 to 5 years in the current role is a point of very high attrition, but as duration increases (10+ years), the rate goes down. What emerges is the pattern that employees in their mid-term tenure are more likely to look elsewhere, but not those with longer tenures, who most could be assumed to have greater satisfaction or perhaps stability of a good job. It could lower your retention rates while solving issues that employees within the 2–5 years range face, and they long for career development opportunities.

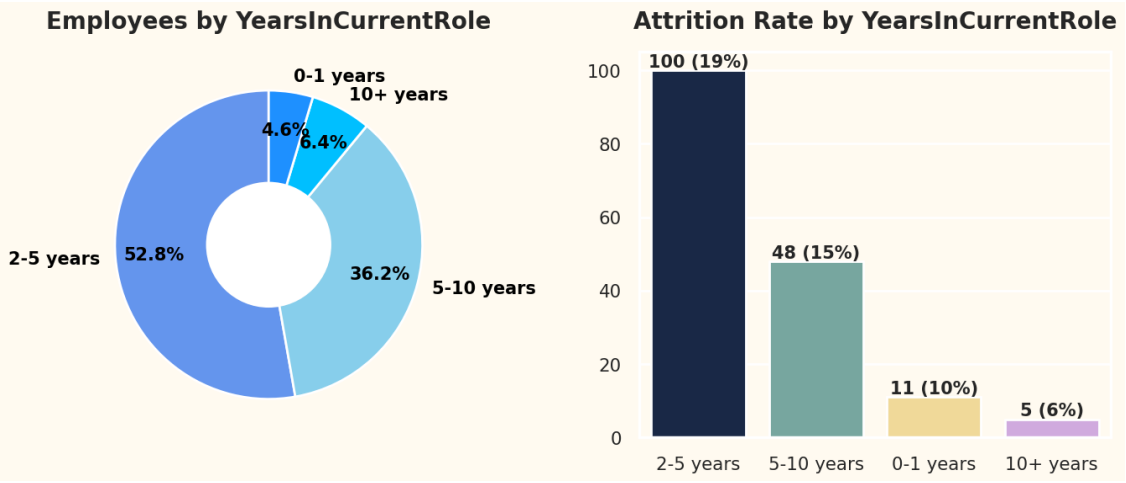


Figure 13.Analysing Employee Attrition by Years In Current Role

14.Analysing Employee Attrition by Years Since Last Promotion.

About 36% of workers have not been promoted in the past two to five years, and nearly eight percent say they haven't seen a promotion for over a decade. Employees with no promotion in the last 5+ years have very high attrition rates throughout all status categories. This points to the reality that limited career progression is one of the major causes of the increased rate of attrition. The organization may need to create more definitive routes for job or career mobility and have a plan review on promotions as strategies to retain employees with higher satisfaction in their work.

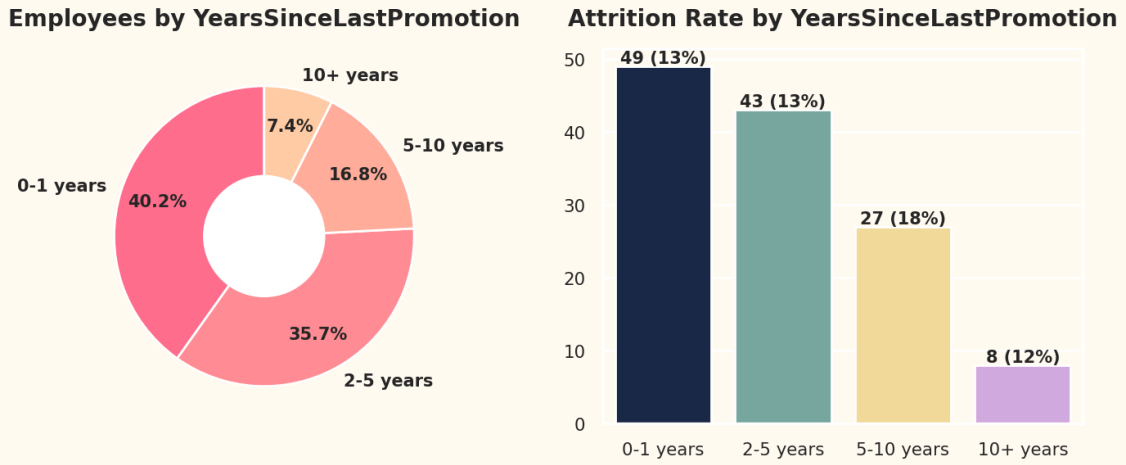


Figure 14.Analysing Employee Attrition by Years Since Last Promotion

15.Analysing Employee Attrition by Years at Company:

Two to ten years are the majority of employees in the organization, with less than one year and over 10 years equally shared by only a few employees. Employees who have been at the company for 2 to 5 years leave more frequently, while those with over a decade of time tend to stay longer. It may indicate that the middle-aged problem is real, and higher levels of dissatisfaction—or inhibiting factors associated with high turnover—affect those more proximal to retirement than they do longer-tenured employees. Addressing the concerns of employees with 2 to 5 years of experience could help ensure smoother retention.

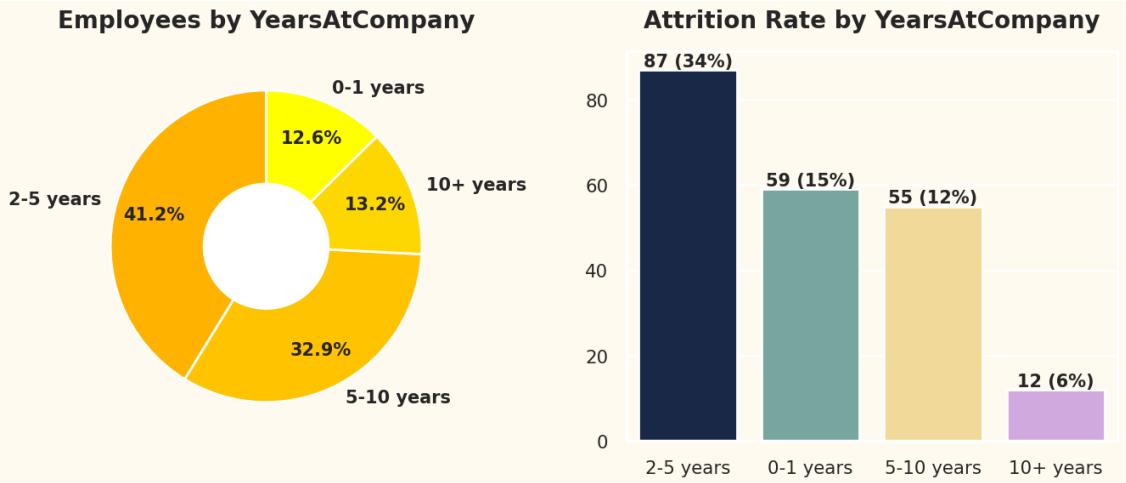


Figure 15.Analysing Employee Attrition by Years at Company

4.2.3 Statistical Analysis - Feature Importance

1. Performing ANOVA Test to Analyse the Numerical Features Importance in Employee Attrition.

Purpose

ANOVA (Analysis of Variance) — This is used to check the effect of multiple numerical features on employee attrition, which is a categorical target column. This test helps us to know whether a numeric variable (like salary, distance from home, etc.) affects significantly the probability of an employee leaving the company.

Statistical Values

The ANOVA test provides two key statistical values:

a) **F-Score :** This value measures the ratio of variance between groups to the variance within groups. A larger F-score indicates a stronger association between the numerical feature and the categorical outcome, suggesting that the numerical feature has a more significant impact on attrition.

b) **P-Value :** This is to quantify the probability of achieving results similar we would conclude that there's no difference . If the p-value of a numerical feature is smaller than the selected significance level (commonly set to 0.05), we can reject the null hypothesis and say that this numerical world significantly influences attrition.

Interpretation

a) **F-Score :** By ranking F-scores, users can see which features will determine whether employees are most likely to leave the organization—higher F-Score = some number is larger just fine.

b) **P-Value :** If the p-value < 0.05, then it means there is a significant relationship between this continuous variable and employee attrition.

By analyzing these values, we can determine which numerical features are most important in influencing employee attrition and thus focus on addressing those areas to improve retention strategies.

2. Visualizing the F_Score of ANOVA Test of Each Numerical features.

The ANOVA test reveals that features such as Age, DailyRate, HourlyRate, MonthlyIncome, MonthlyRate,NumCompaniesWorked,PercentSalaryHike,TotalWorkingYears, TrainingTimesLastYear, YearsAtCompany, and YearsWithCurrManager demonstrate a strong association with employee attrition. This is evidenced by their high F-scores and very low p-values, indicating these factors significantly influence attrition. Conversely, features like DistanceFromHome, StockOptionLevel, YearsInCurrentRole, and YearsSinceLastPromotion show moderate F-scores and extremely high p-values, suggesting they have no significant relationship with attrition. Therefore, it is crucial for the organization to focus on the significant features when devising strategies to reduce attrition rates effectively.

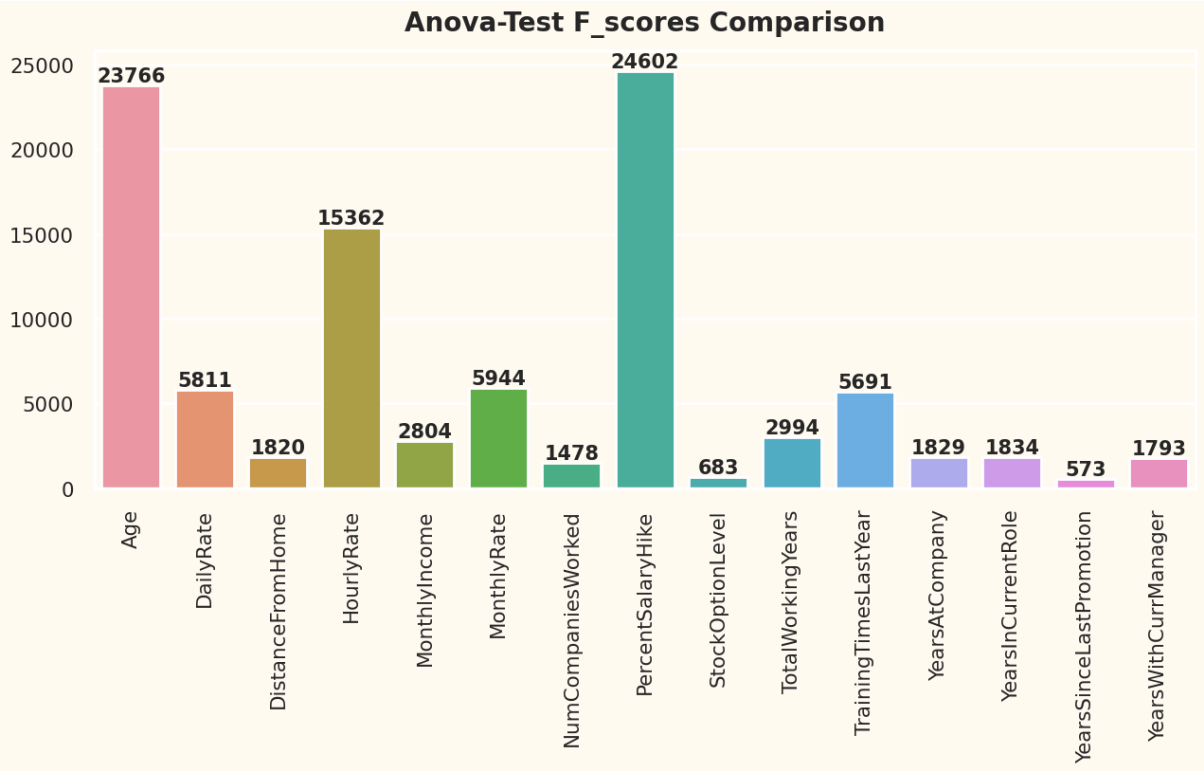


Figure 16.Anova-Test F_scores Comparison

3. Comparing F_Score and P_value of ANOVA Test.

The table below presents the F-scores and p-values for various features analyzed to determine their impact on employee attrition:

Key Insights:

Strong Associations : All features have very low p-values (essentially zero) and high F-scores, indicating a significant impact on attrition. Features such as Age, PercentSalaryHike, and HourlyRate show particularly high F-scores, suggesting a robust association with employee attrition.

Significance : The extremely low p-values across all features indicate that these relationships are statistically significant. This means each feature's influence on attrition is unlikely to be due to random chance.

Actionable Insights : The high F-scores and low p-values suggest that the organization should focus on these features, particularly those with the highest F-scores, when developing strategies to address and reduce employee attrition.

By prioritizing these features, the organization can implement more targeted and effective retention strategies, addressing factors that significantly impact employee turnover.

	Features	F_Score	P_value
0	Age	23766.934042	0.00000000000000000000
1	DailyRate	5811.796569	0.00000000000000000000
2	DistanceFromHome	1820.614585	0.00000000000000000000
3	HourlyRate	15362.122371	0.00000000000000000000
4	MonthlyIncome	2804.459632	0.00000000000000000000
5	MonthlyRate	5944.089071	0.00000000000000000000
6	NumCompaniesWorked	1478.188633	0.00000000000000000000
7	PercentSalaryHike	24602.507947	0.00000000000000000000
8	StockOptionLevel	683.069576	0.00000000000000000000
9	TotalWorkingYears	2994.906310	0.00000000000000000000
10	TrainingTimesLastYear	5691.401732	0.00000000000000000000
11	YearsAtCompany	1829.442766	0.00000000000000000000
12	YearsInCurrentRole	1834.262264	0.00000000000000000000
13	YearsSinceLastPromotion	573.896430	0.00000000000000000000
14	YearsWithCurrManager	1793.291314	0.00000000000000000000

Figure 17.Comparing F_Score and P_value of ANOVA Test

4. Visualizing the Chi-Square Statistic Values of Each Categorical Features.

Chi-Square test was performed for finding the association between categorical features with employee attrition. This is a statistical test applied to determine whether there exists an association between two categorical variables (e.g., department or job role) and whether employees leave or remain in the organization when specific conditions are satisfied.

Categorical Features Selection :

All categorical features in the dataset, excluding the target variable 'Attrition', were selected for analysis.

The Chi-Square test was performed on these tables to obtain the Chi-Square statistic and the p-value. A bar plot was generated to visualize the Chi-Square statistics for each categorical feature, providing a clear representation of their relative strength of association with attrition.

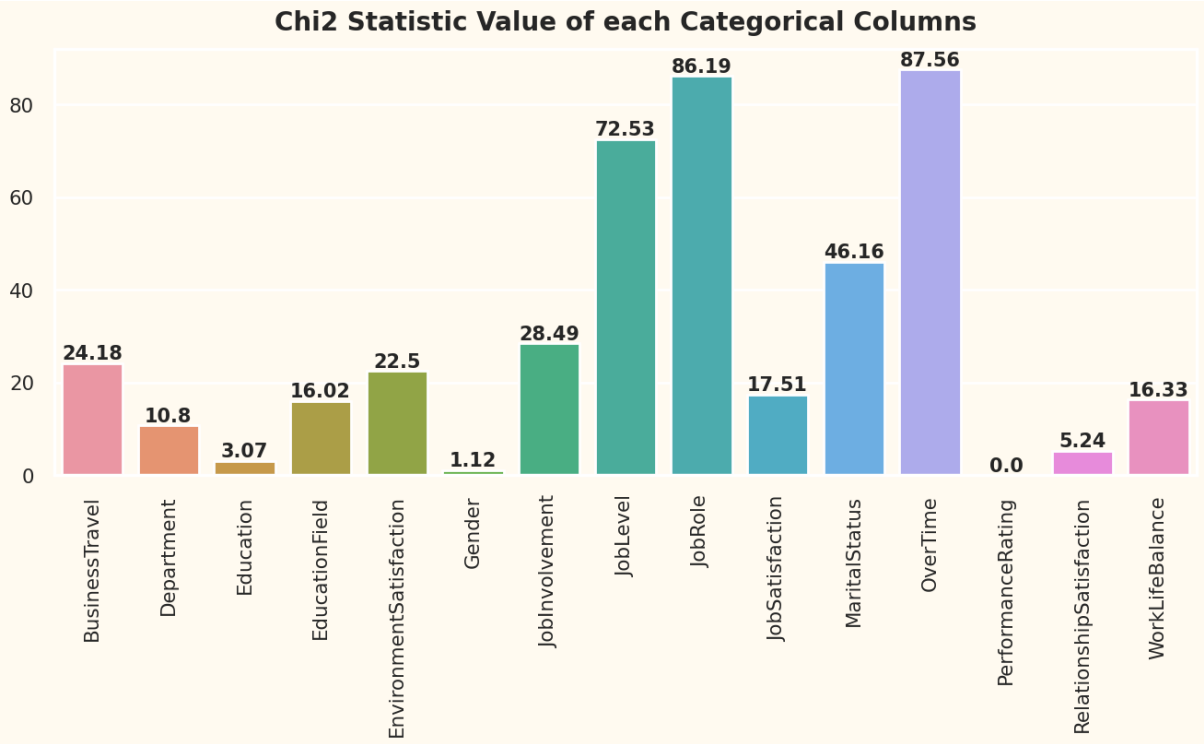


Figure 18.Chi-Square Statistic Values of Each Categorical Columns

5. Comparing Chi2_Statistic and P_value of Chi_Square Test.

Statistically Significant Features : The features with low p-values and high Chi-Square statistics are strongly associated with employee attrition. These include Department, EducationField, EnvironmentSatisfaction, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MaritalStatus, OverTime, and WorkLifeBalance. Their low p-values indicate that the observed associations are unlikely to be due to chance.

Non-Significant Features : Features such as Gender, Education, PerformanceRating, and RelationshipSatisfaction have high p-values and relatively low Chi-Square statistics, suggesting that they do not significantly influence attrition.

While making a strategy for managing employee attrition, the company should concentrate on important features found through the Chi-Square test. By identifying key reasons for why people leave, such as job role and work-life balance, organizations can devise retention strategies that are most likely to be effective. Therefore, the organization can target its intervention by paying more attention to these few drivers, which will improve employee satisfaction on one hand and reduce attrition on the other.

	Features	Chi_2 Statistic	P_value
0	BusinessTravel	24.182414	0.00000560861447644993
1	Department	10.796007	0.00452560657447963286
2	Education	3.073961	0.54552533765659494414
3	EducationField	16.024674	0.00677398013902521211
4	EnvironmentSatisfaction	22.503881	0.00005123468906289433
5	Gender	1.116967	0.29057244902890855265
6	JobInvolvement	28.492021	0.00002863180636713342
7	JobLevel	72.529013	0.00000000000000663468
8	JobRole	86.190254	0.00000000000000275248
9	JobSatisfaction	17.505077	0.00055630045103875563
10	MaritalStatus	46.163677	0.00000000009455511060
11	OverTime	87.564294	0.00000000000000000001
12	PerformanceRating	0.000155	0.99007454659345761616
13	RelationshipSatisfaction	5.241068	0.15497244371052626422
14	WorkLifeBalance	16.325097	0.00097256988453488236

Figure 19.Compairing Chi2_Statistic and P_value of Chi_Square Test

CHAPTER 5. Conclusion

This analysis of employee attrition has revealed several critical insights that can guide organizational strategies aimed at reducing turnover and enhancing workforce stability. By identifying significant associations between various numerical and categorical variables and attrition rates, the findings offer valuable predictors of attrition risk. Nevertheless, the results must be understood in light of certain strictures concerning nature and extent since there may have been other factors affecting employee turnover that were not measured. As such, while these results are robust, they need to be located within a broader context and range of tools for understanding and combating employee turnover.

The analysis highlights several numerical variables that are strongly associated with employee attrition. Age, compensation, and job experience emerged as significant predictors. Older employees appear less likely to leave, potentially due to a greater sense of job security or proximity to retirement. In contrast, younger employees may be more mobile, seeking opportunities that offer faster career advancement or better compensation. Compensation-based variables Monthly Income and Percent Salary Hike are highly important. The smaller the monthly income and wage growth, the more likely employees are to leave if they receive lower pay—this makes sense considering that a competitive salary is the most high-profile competitiveness of job-retention talents. Finally, some job experience variables like Years at Company reveal that employees who have a longer tenure and more experience with the company are also more likely to stay. This suggests that loyalty and a deeper connection to the organization develop over time. Moreover, job-specific variables like Years in the Current Role indicate that employees who have been in the same role for an extended period may either be more stable or at greater risk of leaving if they feel stagnant.

Categorical variables also provided valuable insights into attrition risk. Job-related aspects such as Department, Education Field, Job Role, and Marital Status are associated with attrition. For example, certain departments or job roles may have higher turnover rates due to job demands, workplace culture, or opportunities for career growth. Employees who are single may also be more open to new opportunities over a longer distance from their employer. Factors related to work, such as Job Involvement and Job Satisfaction, are also positively related to employee attrition. Workplaces with less job satisfaction and high dissatisfaction will have employees leave more frequently. This highlights the intricate interaction of job-related and work-related aspects that play a role in reducing employee turnover.

The insights gained from this analysis underscore the importance of a multifaceted approach to reducing employee attrition. Human Resource (HR) strategies should consider both numerical and categorical factors when designing retention programs. For instance, competitive compensation packages that include regular salary hikes can mitigate the risk of losing employees due to financial dissatisfaction. Similarly, providing opportunities for career advancement, particularly for younger or less experienced employees, can help retain this group. Moreover, workplace culture and job satisfaction are equally crucial. Companies should regularly assess employee satisfaction through surveys and feedback mechanisms. These efforts could greatly help reduce turnover, particularly among employees who may feel neither committed to their job nor able to balance life outside of work. For instance, offering flexible hours or even the possibility of remote work as an option could make life easier for employees and improve morale, making them less inclined to seek opportunities elsewhere.

While the analysis offers critical insights, it is important to acknowledge its limitations. The study is based on the available dataset, which may not capture all the variables influencing employee attrition. Unmeasured factors, such as individual employee motivations, external job market conditions, and economic factors, could also play significant roles. Moreover, the data are probably not as representative of actual conditions in different sectors or geographical areas, and therefore, the findings are likely less broadly applicable. To overcome these limitations, future research should include additional variables and account for the external factors that contribute to attrition. Secondly, using qualitative research methods like interviews and focus groups can help us understand more about why experienced staff are leaving. Complementing the quantitative analysis by understanding these nuanced reasons can help create a more effective retention strategy.

In conclusion, this analysis has identified several key numerical and categorical variables associated with employee attrition. Age, compensation, job experience, and specific job-related factors are all significant predictors, as are job-related aspects and work-related factors. These findings highlight the importance of a holistic approach to addressing employee attrition. HR strategies must focus on competitive pay, career development opportunities, and workplace culture improvements. If organizations focus on satisfying their employees both monetarily and non-monetarily, they can create a more engaged workforce, leading to increased loyalty. Yet, it is important to note the study's limitations, and more research needs to be done to understand all factors behind why employees leave. As organizations continually assess and refine their HR strategies, companies can retain their employees, reduce the cost of re-recruiting, and foster a more reliable and productive work environment. Moving forward requires not just implementing what we now know, but also iterating through better approaches based on more research and feedback By doing so, organizations can proactively manage attrition, turning potential challenges into opportunities for growth and improvement.

Recommendations

The report focuses on providing a set of targeted recommendations for curbing employee attrition in the organization. These recommendations stem from important aspects that predict the likelihood of employee turnover, including age, compensation, job tenure, job-related variables, and employment status. Addressing these strategically can help the organization increase employee retention, satisfaction, and overall engagement.

1. **Age-Specific Strategies**
   - **Tailored Development Opportunities**: Roll out specific initiatives to address the individual career needs and ambitions of each age group, including age-specific training for both life skills and technical advancements, and mentoring programs tailored to different generations.
   - **Work-Life Balance Programs**: Provide flexible work arrangement options to meet the needs and performance requirements of a multigenerational workforce, including remote access.

2. **Competitive Compensation**
   - **Compensation Benchmarking**: Utilize HR management system tools to collect data on the latest compensation rates in their field so that they can keep up with industry standards.
   - **Performance-based rewards**: An intertwining of an incentive and pay for performance which helps in encouraging the employees while acknowledging their contributions thereby allowing high-performers to be recognized.

3. **Career Pathing and Job Experience**
   - **Career Pathing**: Create visibility into career progression opportunities, effectively communicating these paths to foster a more engaged and motivated workforce.
   - **Skill Development**: Provide skill development through training programs, workshops, and cross-functional experiences, with regular performance evaluations to nurture growth.

4. **Job-Specific Retention Strategies**
   - **Individualized Retention Plans**: Develop unique strategies to retain employees in various roles and responsibilities. For specialized positions or high-demand roles, focus on increasing job satisfaction and providing challenging assignments aligned with career goals.
   - **positive Work Environment**: Support and Encouragement: Create an atmosphere that supports, encourages people to talk about their achievements, and addresses the concerns of employees in their job.

5. **Enhancing Job-Related Aspects**
   - **Employee Engagement**: Foster a supportive work culture that encourages professional growth and lifelong learning, ensuring transparent and fair processes for promotions and career growth.
   - **Work-Life Balance**: Promote work-life balance by managing scheduling requirements, offering flexible schedules, and supporting staff wellness initiatives.

6. **Workplace Factors**
   - **Increase Environment Satisfaction**: Regularly assess workplace satisfaction through employee surveys and act on the feedback. Address issues related to job involvement, work-life balance, and employee satisfaction proactively.
   - **Feedback Loops**: Establish strong avenues for employees to share their concerns and ideas, tracking feedback and making changes as needed.

7. **Overall Organizational Culture**
    - **Positive organizational culture**: one where employees are supported, rather than made to feel isolated, and where everyone has access to the help and resources they need to thrive.
    - **Continuous Improvement**: Revisit retention strategies often to realign with changing employee needs and organizational objectives, taking feedback regularly.

By following these suggestions, not only will you have much lower employee turnover rates but also improve the satisfaction of your overall staff and their engagement in work (and hence production). Focusing on a range of strategies that meet the needs of its heterogeneous workforce will both attract and retain talent sustainable success for your company.

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Appendices

STT	Attribute Name	Meaning
1	Age	Employee's age
2	Gender	Employee's Gender
3	BusinessTravel	Frequency of employees' business trips
4	DailyRate	Daily salary rate for employees
5	Department	Office of employees
6	DistanceFromHome	Distance from home in miles to work
7	Education	Level of education achieved by staff
8	EducationField	Employee's field of study
9	EmployeeCount	Total number of employees in the organization
10	EmployeeNumber	A unique identifier for each employee record
11	EnvironmentSatisfaction	Employee satisfaction with their working environment
12	HourlyRate	Hourly rate for employees
13	JobInvolvement	Level of involvement required for the employee's job
14	JobLevel	Employee's level of work
15	JobRole	The role of employees in the organization
16	JobSatisfaction	Employee satisfaction with their work
17	MaritalStatus	Employee's marital status
18	MonthlyIncome	Employee's monthly income
19	MonthlyRate	Monthly salary rate for employees
20	NumCompaniesWorked	Number of companies the employee worked for
21	Over18	Whether the employee is over 18 years old
22	OverTime	Do employees work overtime
23	PercentSalaryHike	Salary increase rate for employees
24	PerformanceRating	The performance rating of the employee
25	RelationshipSatisfaction	Employee satisfaction with their relationships
26	StandardHours	Standard working hours for employees
27	StockOptionLevel	Employee stock option level
28	TotalWorkingYears	Total number of years the employee has worked
29	TrainingTimesLastYear	Number of times employees were taken to training in the last year
30	WorkLifeBalance	Employees' perception of their work-life balance
31	YearsAtCompany	Number of years employees have been with the company
32	YearsInCurrentRole	Number of years the employee has been in their current role
33	YearsSinceLastPromotion	Number of years since employee's last promotion
34	YearsWithCurrManager	Number of years an employee has been with their current manager
35	Attrition	Does the employee leave the organization

Figure 20.Dataset Features Description

	Total No. of Missing Values	% of Missing Values
Age	0	0.0
Attrition	0	0.0
BusinessTravel	0	0.0
DailyRate	0	0.0
Department	0	0.0
DistanceFromHome	0	0.0
Education	0	0.0
EducationField	0	0.0
EmployeeCount	0	0.0
EmployeeNumber	0	0.0
EnvironmentSatisfaction	0	0.0
Gender	0	0.0
HourlyRate	0	0.0
JobInvolvement	0	0.0
JobLevel	0	0.0
JobRole	0	0.0
JobSatisfaction	0	0.0
MaritalStatus	0	0.0
MonthlyIncome	0	0.0
MonthlyRate	0	0.0
NumCompaniesWorked	0	0.0
Over18	0	0.0
OverTime	0	0.0
PercentSalaryHike	0	0.0
PerformanceRating	0	0.0
RelationshipSatisfaction	0	0.0
StandardHours	0	0.0
StockOptionLevel	0	0.0
TotalWorkingYears	0	0.0
TrainingTimesLastYear	0	0.0
WorkLifeBalance	0	0.0
YearsAtCompany	0	0.0
YearsInCurrentRole	0	0.0
YearsSinceLastPromotion	0	0.0
YearsWithCurrManager	0	0.0

Figure 21.Total No. of Missing Values and the Percentage of Missing Values

	count	mean	std	min	25%	50%	75%	max
Age	1470.0	36.923810	9.135373	18.0	30.00	36.0	43.00	60.0
DailyRate	1470.0	802.485714	403.509100	102.0	465.00	802.0	1157.00	1499.0
DistanceFromHome	1470.0	9.192517	8.106864	1.0	2.00	7.0	14.00	29.0
EmployeeCount	1470.0	1.000000	0.000000	1.0	1.00	1.0	1.00	1.0
EmployeeNumber	1470.0	1024.865306	602.024335	1.0	491.25	1020.5	1555.75	2068.0
HourlyRate	1470.0	65.891156	20.329428	30.0	48.00	66.0	83.75	100.0
MonthlyIncome	1470.0	6502.931293	4707.956783	1009.0	2911.00	4919.0	8379.00	19999.0
MonthlyRate	1470.0	14313.103401	7117.786044	2094.0	8047.00	14235.5	20461.50	26999.0
NumCompaniesWorked	1470.0	2.693197	2.498009	0.0	1.00	2.0	4.00	9.0
PercentSalaryHike	1470.0	15.209524	3.659938	11.0	12.00	14.0	18.00	25.0
StandardHours	1470.0	80.000000	0.000000	80.0	80.00	80.0	80.00	80.0
StockOptionLevel	1470.0	0.793878	0.852077	0.0	0.00	1.0	1.00	3.0
TotalWorkingYears	1470.0	11.279592	7.780782	0.0	6.00	10.0	15.00	40.0
TrainingTimesLastYear	1470.0	2.799320	1.289271	0.0	2.00	3.0	3.00	6.0
YearsAtCompany	1470.0	7.008163	6.126525	0.0	3.00	5.0	9.00	40.0
YearsInCurrentRole	1470.0	4.229252	3.623137	0.0	2.00	3.0	7.00	18.0
YearsSinceLastPromotion	1470.0	2.187755	3.222430	0.0	0.00	1.0	3.00	15.0
YearsWithCurrManager	1470.0	4.123129	3.568136	0.0	2.00	3.0	7.00	17.0

Figure 22.Performing Descriptive Analysis on Numerical Attributes

	count	unique	top	freq
Attrition	1470	2	No	1233
BusinessTravel	1470	3	Travel_Rarely	1043
Department	1470	3	Research & Development	961
Education	1470	5	Bachelor	572
EducationField	1470	6	Life Sciences	606
EnvironmentSatisfaction	1470	4	High	453
Gender	1470	2	Male	882
JobInvolvement	1470	4	High	868
JobLevel	1470	5	Entry Level	543
JobRole	1470	9	Sales Executive	326
JobSatisfaction	1470	4	Very High	459
MaritalStatus	1470	3	Married	673
OverTime	1470	2	No	1054
PerformanceRating	1470	2	Excellent	1244
RelationshipSatisfaction	1470	4	High	459
WorkLifeBalance	1470	4	Better	893

Figure 23.Performing Descriptive Analysis on Categorical Attributes

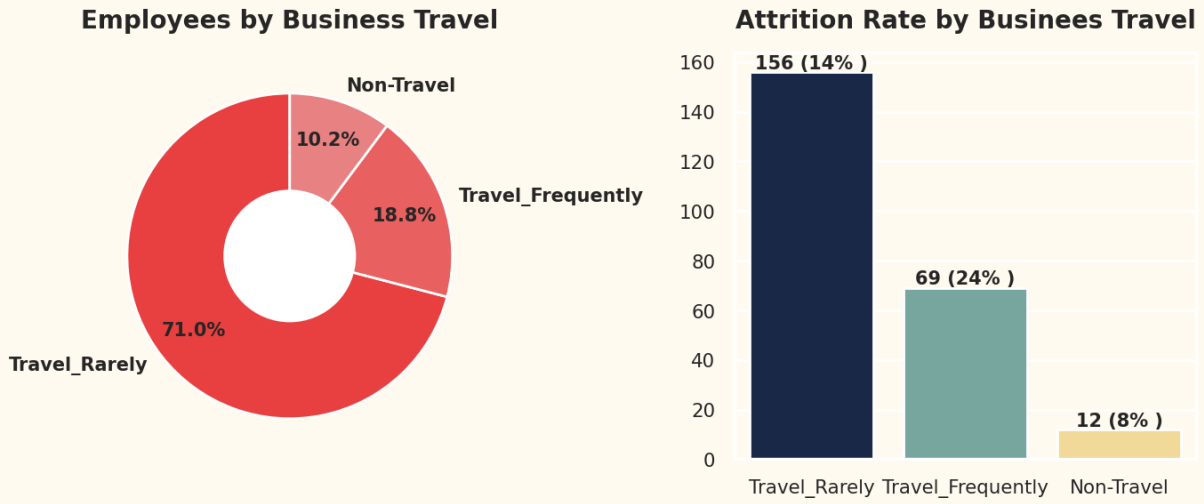


Figure 24.Analyzing Employee Attrition by Business Travel

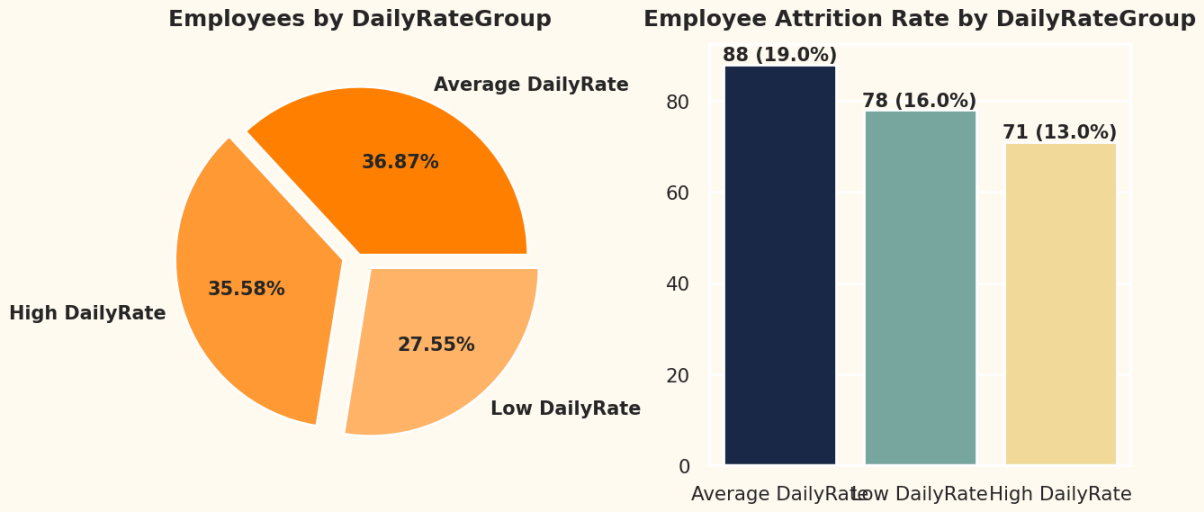


Figure 25.Analyzing Employee Attrition by DailyRate

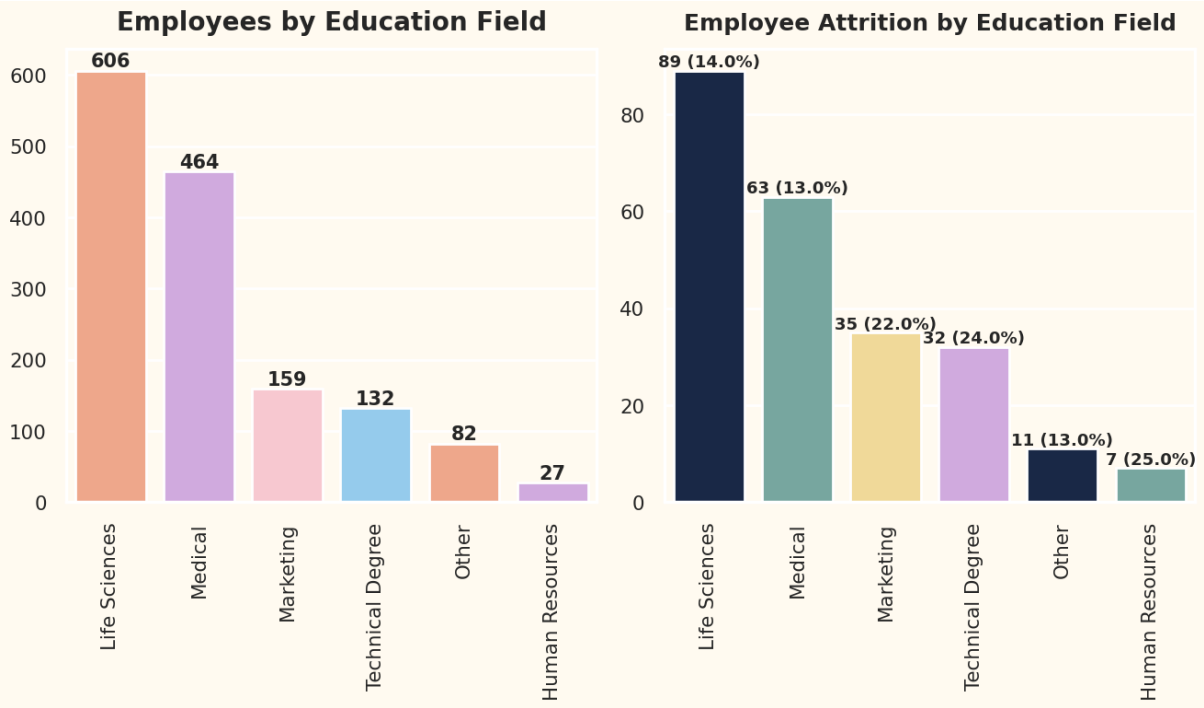


Figure 26.Analyzing Employee Attrition by Education Field

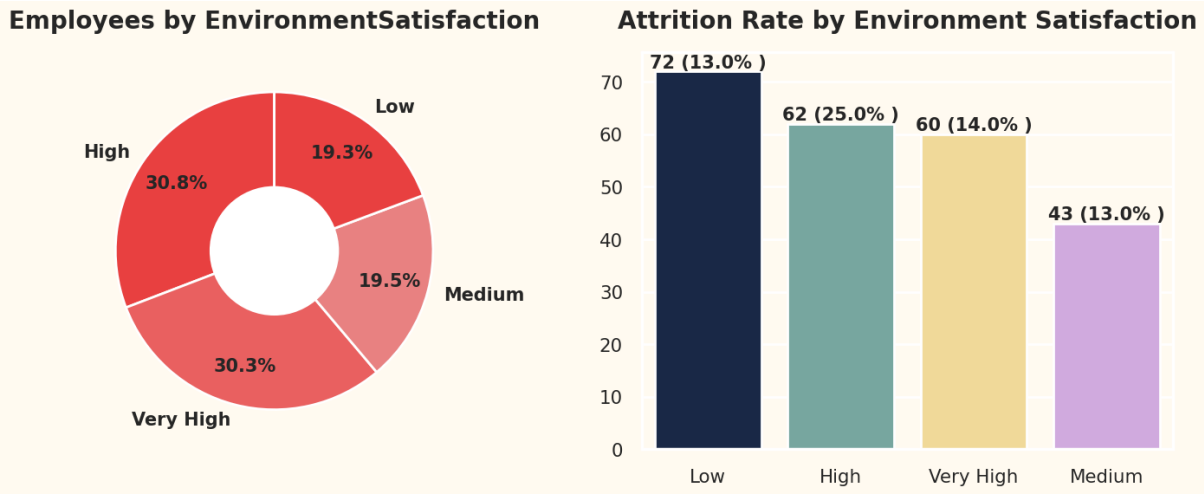


Figure 27.Analyzing Employee Attrition by Environment Satisfaction

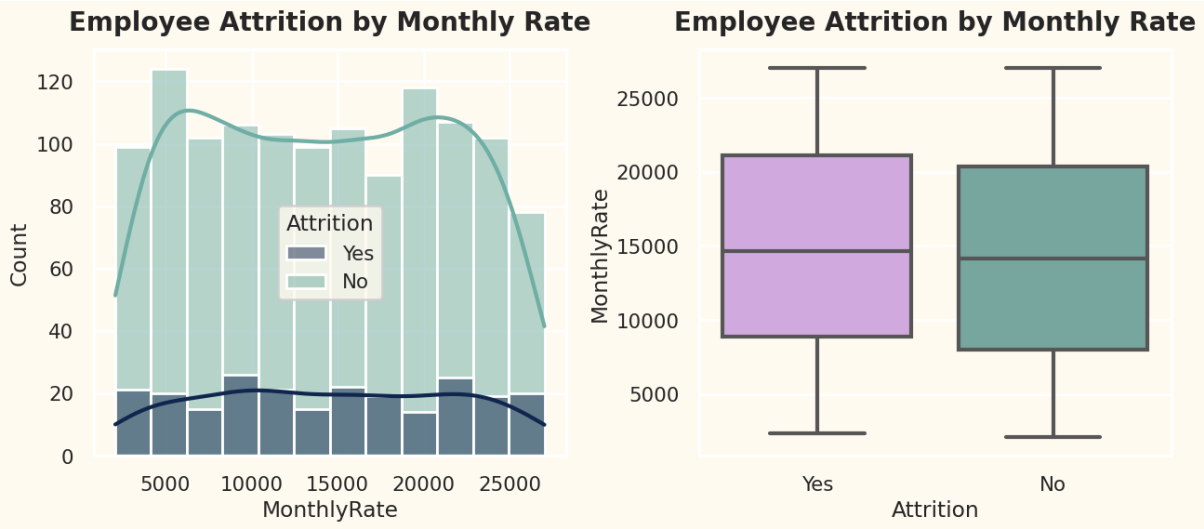


Figure 28.Analyzing Employee Attrition by Monthly Rate

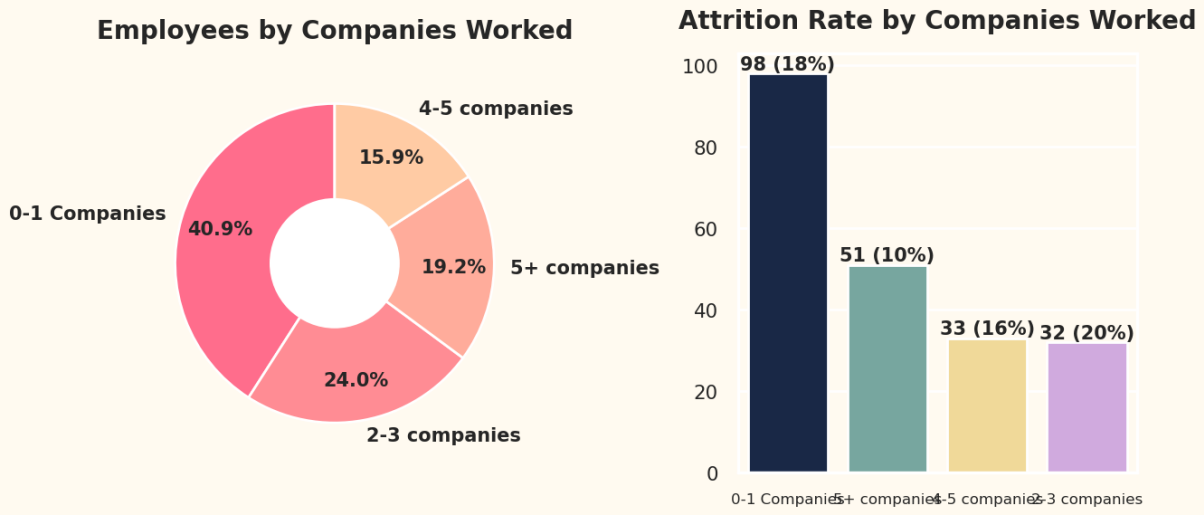


Figure 29.Analyzing Employee Attrition by Number of Companies Worked

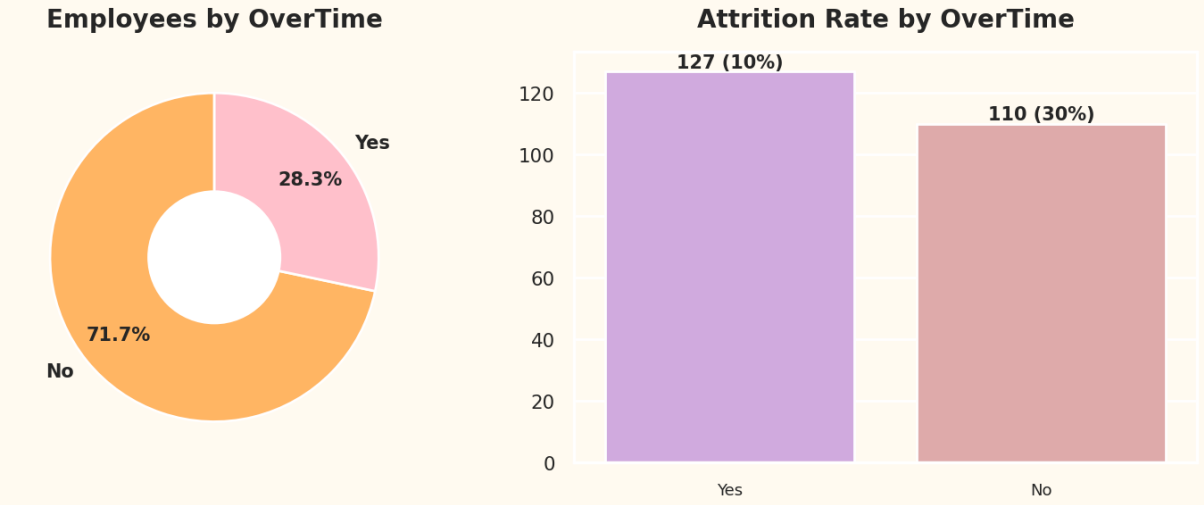


Figure 30.Analyzing Employee Attrition by Over Time

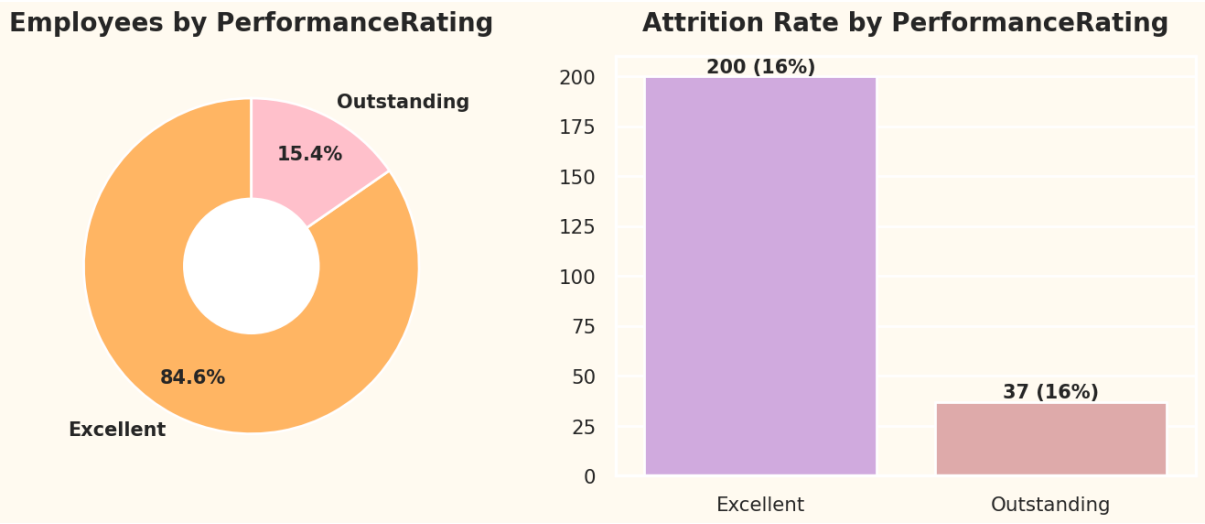


Figure 31.Analyzing Employee Attrition by Performance Rating

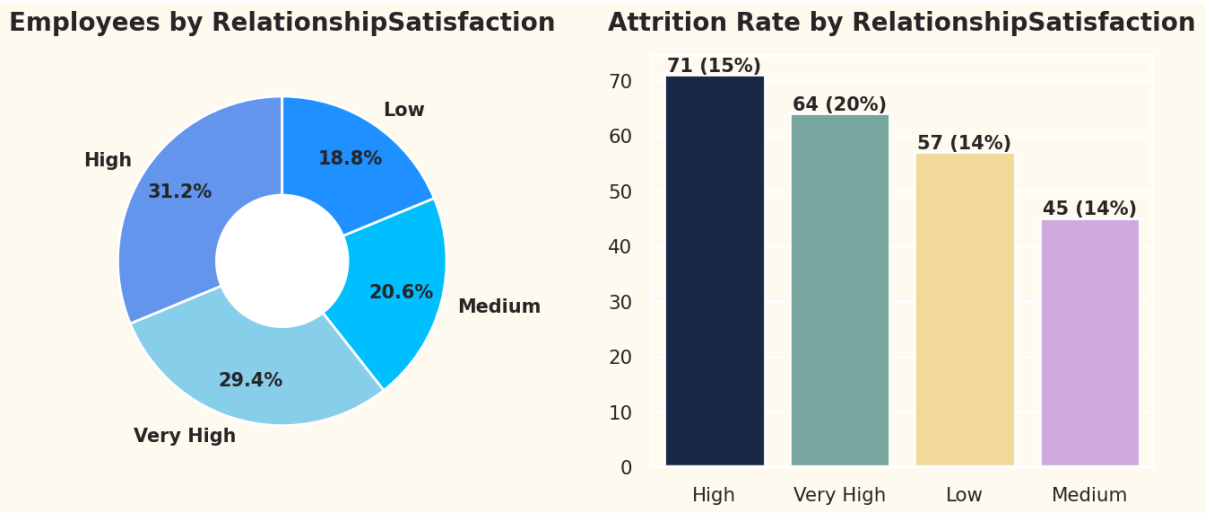


Figure 32.Analyzing Employee Attrition by Relationship Satisfaction

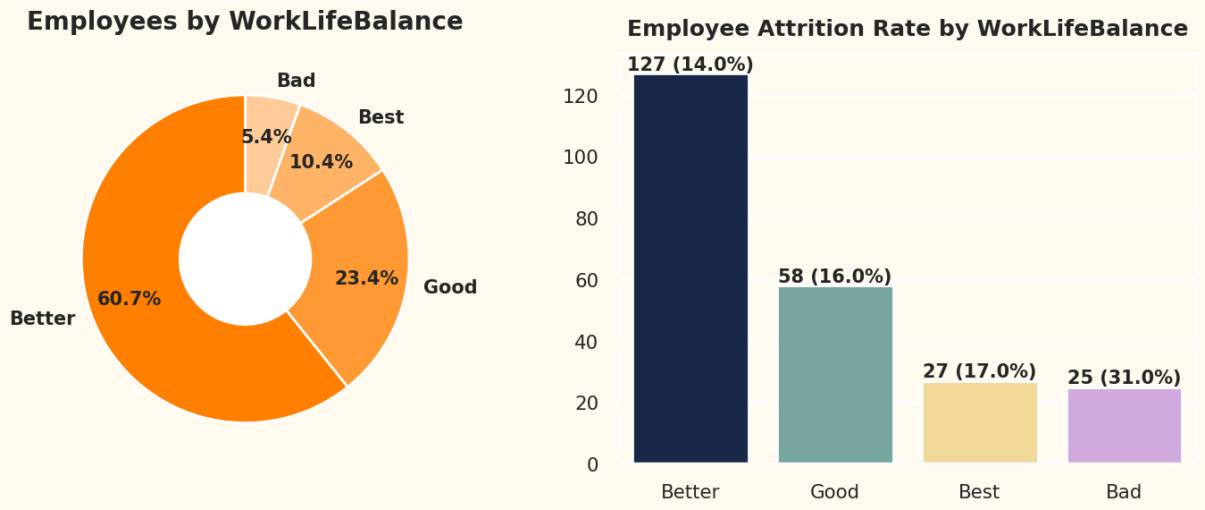


Figure 33.Analyzing Employee Attrition by Work Life Balance

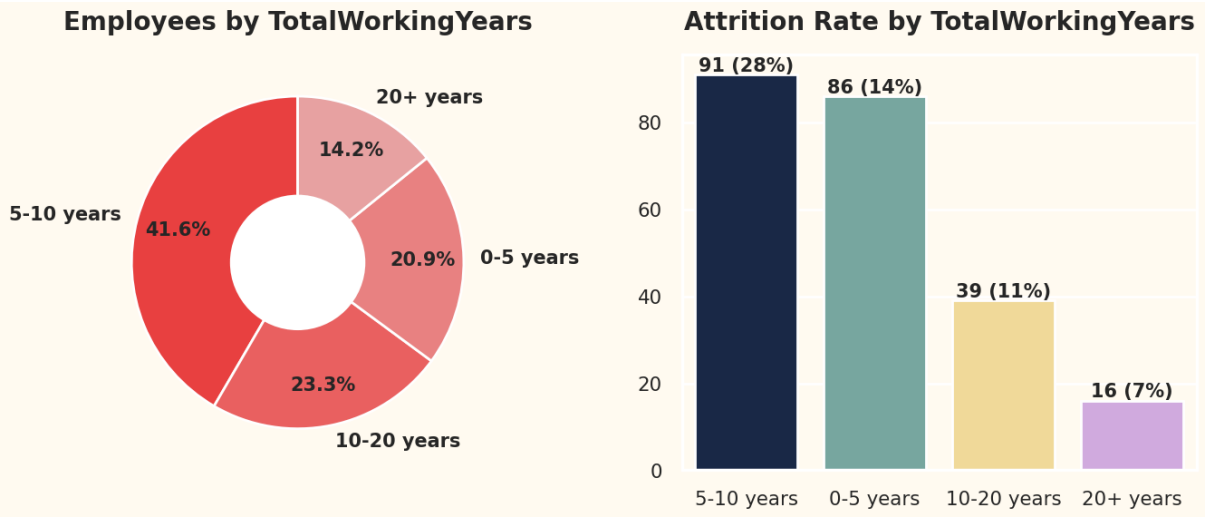


Figure 34.Analyzing Employee Attrition by Total Working Years

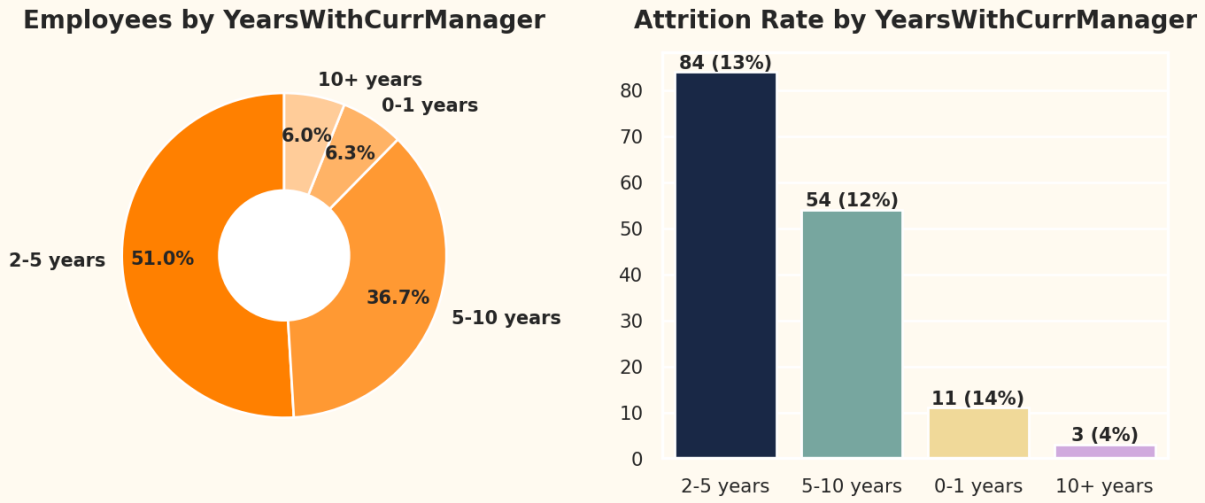


Figure 35.Analyzing Employee Attrition by Years with Current Manager